

PyPSA-Earth sector-coupled: A global open-source multi-energy system model showcased for hydrogen applications in countries of the Global South

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ABSTRACT

This study presents sector-coupled PyPSA-Earth: a novel global open-source energy system optimization model that incorporates major demand sectors and energy carriers in high spatial and temporal resolution, to enable energy transition studies worldwide. The model includes a workflow that automatically downloads and processes the necessary demand, supply and transmission data to co-optimize investment and operation of energy systems of countries or regions of Earth. The workflow provides the user with tools to forecast future demand scenarios and allows for custom user-defined data in several aspects. Sector-coupled PyPSA-Earth introduces novelty by offering users a comprehensive methodology to generate readily available sector-coupled data and model of any region worldwide, starting from raw and open data sources. The model provides flexibility in terms of spatial and temporal detail, allowing the user to tailor it to their specific needs. The capabilities of the model are demonstrated through two showcases for Egypt and Brazil. The Egypt case quantifies the relevant role of PV, exceeding 35 GW, and electrolysis in Suez and Damietta regions, for meeting 16% of the EU hydrogen demand. Complementarily, the Brazil case confirms the model's ability in handling hydrogen planning infrastructure, including repurposing of existing gas networks which results in 146 M€ lower costs than building new pipelines. The results prove the suitability of sector-coupled PyPSA-Earth to meet the needs of policymakers, developers, and scholars in advancing the energy transition. The authors invite the interested individuals and institutions to collaborate in the future developments of the model within PyPSA meets Earth initiative.

1. Introduction

In the face of escalating climate change impacts, the global imperative to mitigate greenhouse gas emissions has intensified, necessitating a transition towards a renewable energy-based system. This transition requires a holistic approach that encompasses the entire world economy accounting for the intertwined synergies of all its sectors. The scientific community is increasingly recognizing that conventional paradigms, which focus solely on electricity sector-based energy system

planning, are inadequate as they ignore carbon emissions from other sectors and neglect significant flexibility potentials. Sector coupling, on the other hand, creates various flexibility options that can smoothen the supply–demand mismatches, for example, through power-to-X technologies that converts surplus electricity to storable fuels like hydrogen. In addition, integrating the different sectors allow for the displacement of fossil fuels with renewable electricity or synthetic fuels across the traditionally hard-to-abate sectors like industry and transport. Finally,

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it allows for cost-effective infrastructure expansion by integrating different energy carriers and technologies reducing the need for extensive transmission grid expansion when combined with gas infrastructure or cross-carrier storage solutions [1–4].

While there are many examples of applied sector-coupled energy system models on the national, EU [5] and global level [6], no global open-source sector-coupled multi-energy system model exists that uses high spatial and temporal resolved open data – meaning data in hourly and sub-national resolution. Sector-coupled PyPSA-Earth presented in this paper fills the gap. Many applicable sector-coupled models are either not open-source or require license fees for the underlying programming language (e.g. GAMS), which excludes many research and industry stakeholders from accessing, tailoring and testing the tools. This category includes prominent and powerful models built with EnergyPlan [7], TIMES [8], PLEXOS [9], or the LUT model [10]. Furthermore, many of these tools, as well as open-source alternatives such as GENeSYS-MOD [11], NEMO [12], LOADMATCH [13] and H2RES [14] often disregard high-resolution spatial data, which are necessary to account for transmission bottlenecks, or disregard hourly time resolutions, which leads to misrepresent the flexibility needs and contemporaneity of renewable sources [1,15]. Further comparison between the different modelling frameworks and their specific features is provided in the previous study [16].

While there seems to be no suitable open and accessible solutions on a global scale, some evolved on a continental level. In this context, PyPSA-Eur [17] and its predecessors, PyPSA-Eur-Sec [5] and PyPSA-Eur-Sec-30, represent critical milestones. These models were pioneers in the integration of sector coupling and regional grid integration with sub-national spatial resolution and hourly time resolution over an entire year, thereby enabling a more nuanced and comprehensive analysis of the European energy system [3]. Building on these foundational works, the PyPSA meets Earth team was first established to expand the effort first to Africa and then the entire world [16]. Towards this goal, the PyPSA-Earth model was created, making it the first global open power-system model with high spatial and temporal data that can be customized for any country or continent [16,18]. However, achieving the ambitious goal of a global energy transition requires ongoing innovation and refinement of modelling tools, particularly the extension of sector-coupled capabilities, akin to the pioneering ongoing work.

In this vein, sector-coupled PyPSA-Earth presented in this paper represents the latest advancement, incorporating sector-coupling to extend PyPSA-Earth [16] from a power-only model to a multi-energy-system model at a global scale. This is achieved by expanding the optimization problem to incorporate multiple energy carriers, enabling conversions between the different types and the flow across the network. In addition, the model now integrates various demand sectors beyond the residential sector such as the industry and transport. Furthermore, the model incorporates policy mechanisms focusing on hydrogen as a key fuel for both domestic use as well as exports. The study introduces the following novelty list:

- (i) The first open-source sector-coupled energy system model with global coverage and high resolution.
- (ii) Embedded projection tool for residential, transport, industry, agriculture, and services demand.
- (iii) An automatically compiled geo-asset database of demand capacities sourced from open-data.
- (iv) Flexible integration of custom datasets, when available and preferred by the user.

The structure of the upcoming sections is divided as follows: Section 2 introduces the model workflow divided into three components — demand, supply, and transmission. Section 3 highlights the model's flexibility while Section 4 focuses on the hydrogen framework. Section 5 explains the mathematical formulation of the model and Section 6 addresses the data validation process. Two practical showcases

are presented in Section 7 to demonstrate the model's capabilities. Finally, Section 8 concludes the study and suggests key topics for further studies.

2. Model workflow

2.1. Description

Sector-coupled PyPSA-Earth is a readily available open source model that allows researchers to transparently analyse and plan the decarbonized integrated energy system of the future of any country or region. The model contains the workflow shown in Fig. 1 that leverages on open datasets to tailor the model to the country or region of choice and obtain complete optimization results for policy, technical, economic and environmental impact. The model has flexible spatial and temporal resolution, allowing the user to tune it to their specific application.

The main functionality of sector-coupled PyPSA-Earth is the co-optimization of investment and dispatch of multiple energy carrier technologies to serve various demand sectors at minimum system cost within a certain carbon emission limit. The demand sectors include residential, industry, transport (including road, rail, aviation and navigation), agriculture, and services. In addition to electricity, the model considers the following energy carriers: hydrogen, natural gas, biomass, oil, and heat, as well as carbon dioxide in the form of emissions and feed-stock for the synthesis of other carriers. Existing energy infrastructures such as power grid, gas networks as well as generation assets are assessed and accounted for in the model on a transmission level; moreover, their capacity expansion is also supported. The model includes storage technologies and the synthesis of energy carriers, which give the tool ample flexibility options for short- and long-term system dispatch. The former includes pumped hydro storage (PHS), battery electric vehicles (BEV), heat storage in water tanks and hydrogen and carbon dioxide storage, while the latter covers power-to-liquid and power-to-gas technologies. The model utilizes the PyPSA Technology Data repository [19] for default model assumptions of the different technologies and energy carriers.

The model supports a large spectrum of sector-coupling studies. Examples of this include: long-term planning of generation and transmission needs for an integrated system, optimal roll out of power-to-X technologies, cost analysis of hydrogen and synthetic fuels production, repurposing of natural gas networks for hydrogen use, electrification of the road transport and the industry sectors, and assessment of long-term cross carrier storage options, among others. Two example cases are presented in this paper to demonstrate the suitability of sector-coupled PyPSA-Earth for energy transition studies.

2.2. Demand modelling

Demand projection is one key block for accurate energy system modelling. Its estimation depends on various socio-economic, environmental and political factors that are complex to account for, thus a robust but simplified approach is needed. Sector-coupled PyPSA-Earth provides a complete demand data workflow that supports every country in the world allowing for an effortless modelling of the energy system. The demand data preparation is a twofold process that automatically fetches multiple datasets from open data sources. The first step is preparing the national-level data for the different sectors of the countries of interest. Then, in a second step, the aggregated regional demand is distributed geospatially accounting for the locations of the main demand centers for each sector, that is an essential step for high spatial resolution modelling. For example, the main demand centers for aviation are the airports and for navigation are ports. As for the temporal profiles, they differ per sector. The temporal profiles for residential and services demand are estimated using the Global Energy GIS tool [16]. Based on socio-economic predictors, such as population, gross domestic product, and weather information, the tool generates

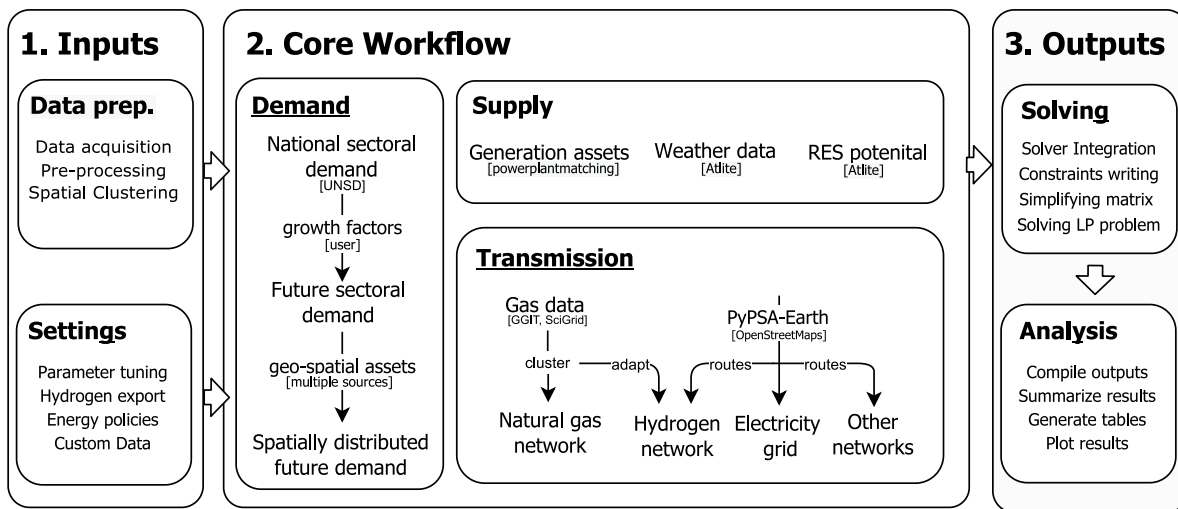


Fig. 1. Schematic representation sector-coupled PyPSA-Earth workflow. The diagram depicts the three main parts of the workflow, starting by (1) Inputs, which involves fetching the necessary data as well fine tuning the model parameters. Secondly, (2) Core Workflow which includes the main data processing for the demand, supply and transmission. Finally (3) Outputs which covers the LP problem preparation and solving as well as generating analysis figures and tables.

hourly demand time series proven to be trained in good agreement with historical observation [20]. The heat demand in both sectors relies on Atlite [21] to generate a daily heat profile based on the daily average temperature of each node in addition to a standard intra-day profile from the German Association of Energy and Water Industries (Bundesverband der Energie- und Wasserwirtschaft, BDEW) [22] that differentiates between weekdays and weekends. The road transport demand profile utilizes traffic data by the German Federal Highway Research Institute (Bundesanstalt für Straßenwesen, BAST) [23]. The remaining demand sectors are by default constant, but can be adjusted when custom profiles are available.

2.2.1. National demand data

In the first step, the Energy Statistics from the United Nations Statistics Division UNSD [24] is used as the default source for the national level demand data. The workflow extracts the final energy demand of the following demand sectors from the full energy balance dataset: residential households, road transport, rail transport, domestic and international aviation, domestic and international navigation, services, agriculture, as well as 12 different industries. The latter comprises Iron and steel, Chemical and petro-chemical, Non-ferrous metals, Non-metallic minerals, Transport equipment, Machinery, Mining and quarrying, Food and tobacco, Paper, pulp and printing, Wood and wood products, Textile and leather, and Construction. The data incorporates the specific energy carrier demand for each of these sectors on a national level. The process is executed automatically with the model run by downloading and post-processing the data. The post-processing stage involves filtering the dataset based on the user-selected country or region and the specified base year, in addition to aggregating the specific fuels used in the sector to their corresponding representative carriers in the model. This aggregation employs energy conversion factors from the supplementary material provided by UNSD [25] to map the different fossil fuels to the representative fuel considered in the model. For instance Kerosene, Naphtha and Ethane demand is converted using the energy content to the representative fuel: Oil. Lignite, hard coal and coal tart are included within the representative fuel Coal. Biogas, fuel wood and waste are included within Biomass. This process creates the base energy demand dataset that is used as the starting point of the future demand scenario projections. Projecting the future energy demand of the different sectors is carried out by making use of a set of parameters, the user can adapt to create different future scenarios. The parameters include an activity growth factor in the form of compound annual growth rate (CAGR), efficiency gains

factors in the same form and an optional hard fuel shift factor to project future electrification of the different sectors. Fine-tuning the keys allows the user to create various future demand scenarios for the different time horizons resulting in the desired projections of sectoral final energy demand for the target year of study. Fig. 2 shows the sectoral demand of the base year of 2019 of the Oceania region and an example projection onto years 2030, 2040 and 2050. The scenario presented is a moderately high electrification scenario with an overall large efficiency gains especially in the transport sector, and no hard fuel shifts. Full parameter list is presented in Tables A.1, A.2, and A.3 in the annex. While the model allows each of the parameters of the list to be set to each country independently, for simplicity in this scenario, the parameters are unified over all the countries of Oceania.

2.2.2. Geo-spatial assets

In the second step, the workflow creates a global geospatial database that determines the sub-national distribution of the demand across the nodes of the system; examples of results are depicted in Fig. 3. Datasets with global coverage are used to facilitate the distribution of the national level demand on the different locations of the system. The industry database comprises the following: iron and steel, with 3 different production technologies from the Global Steel Plant Tracker [26], cement production assets through the Global Database of Cement Production Assets [27] by the Spatial Finance Initiative, petrochemical and chemical industries through Petrochemicals Database for Europe and North America [28] and Global Oil Refinery Complex and Daily Capacity Dataset [29], Mining through the open database on global coal and metal mine production [30], and finally Paper and pulp production locations in South America from the Spatial Finance Initiative [31]. Fig. 3(a) depicts a selected sample of the geo-referenced industry database created to cover the most energy intensive industries. In addition, to distribute the aviation and navigation demand, an airport and port database with global coverage is compiled from the global dataset of airports by OurAirports [32] and the World Port Index [33]; an example subset for Africa is shown in Fig. 3(b). More figures for the other continents can be found in the Annex in Figs. A.1–A.12. The capacities of the industrial facilities and the sizes of the airports and ports are used as the distribution keys to disaggregate the national demand to the different nodes as explained in Section 5.2.2. When no geo-referenced data is available, the demand of that industry is distributed according to the gross domestic product (GDP) index. For the residential, services, road and rail transport, the population density dataset is used as distribution keys.

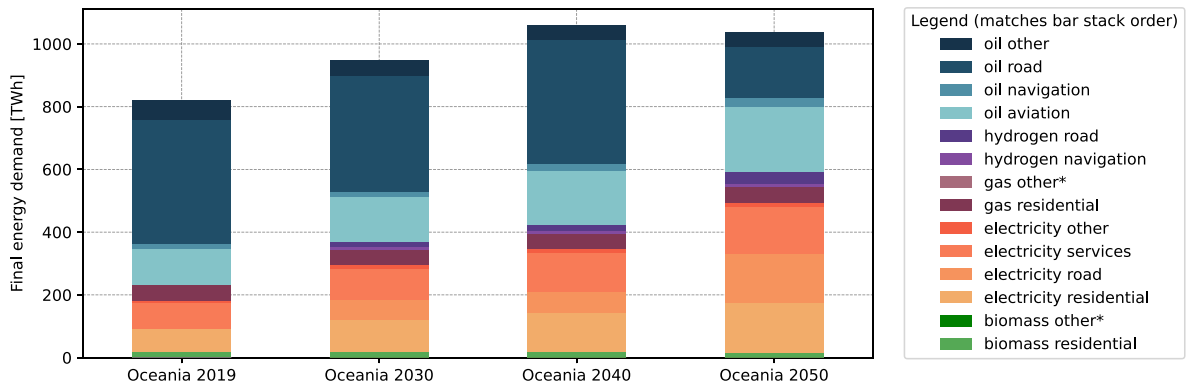


Fig. 2. Demand projection of the countries in Oceania for the short-, medium- and long-term time horizons shown for the residential, transport, services and agriculture sectors. The base-case demand year of 2019 is used to project the sectoral demand of the years 2030, 2040, and 2050. The legend is ordered as the stack sequence: “other” shows all other sectors combined, and the carrier-sector combination annotated with * are not visible on the figure. Oil and gas can be derived from natural sources or created synthetically, this is optimized by the model according to the allowed emission cap. The parameters used for this projection are presented in [Tables A.1–A.2](#) in the [Appendix](#).

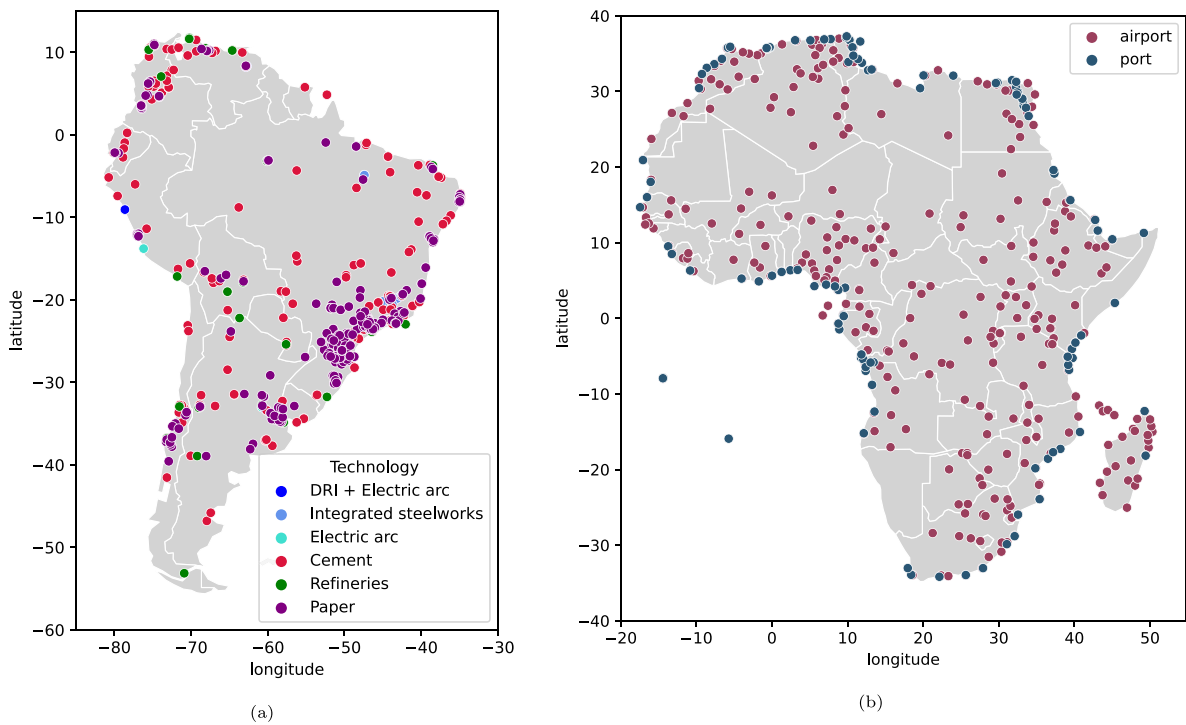


Fig. 3. Location of major industries, airports, and ports available in the database of sector-coupled PyPSA-Earth. The data is used mainly to disaggregate the national-level demand data into the different regions of the countries under study. (a) Geographical distribution of the major industries in South America broken down by commodity and technology. (b) Geographical distribution of the airports and ports in Africa.

2.3. Supply modelling

The supply data is obtained by combining multiple datasets of two major categories: renewable energy sources and fossil-fuel energy sources. The methodology heavily relies on the original PyPSA-Earth formulation [16].

2.3.1. Renewable energy sources

Beyond costs parameters, the major critical information for RES is the specific time series of energy production. Except for few exceptions, e.g. geothermal, the specific production is highly dependent on the weather conditions that highly fluctuate within the year and over long periods of time. To successfully capture these capabilities and draw time series profile for arbitrary weather conditions and a variety of technologies, the Atlite package [21] has been deemed useful for the study, akin to PyPSA-Earth [16]. Atlite successfully enables collecting

weather information and transform them into power time series for energy production, by using well-established technical models of selected RES technologies. Atlite not only supports electricity generating technologies, such as varieties of photovoltaic modules or wind turbines, but also other energy vector thermal, e.g. solar thermal.

To achieve this, Atlite performs the following steps. First, it generates defined spatio-temporal boundaries, known as cutouts. Then, these cutouts are populated with environmental and weather data, drawn from various datasets such as ERA5 reanalysis data [34], SARAH-2 satellite data [35], and GEMCO bathymetry [36]. Finally, conversion functions are applied to create technology-specific, spatially detailed time series and potentials [21]. The conversion functions are based on established models that capture the relationship between the production time series and the determinants of the production for each technology, e.g. irradiance and ambient temperature for photovoltaic energy or solar thermal production.

The supported RES technologies are solar photovoltaic, on- and offshore wind turbines, geothermal as well as hydro-runoff, reservoir, and dam power resources. For hydropower technology, a normalization procedure is performed to match the overall energy production of existing power plants by country [37].

2.3.2. Existing power plants

Power plants are the backbone of existing energy systems, and they are long-lasting investments spanning multiple decades. Accordingly, having their accurate representation is essential to plan for future energy systems. Various private, national and public datasets are available for estimating the installed capacity. However, the exact locations of the powerplants are rarely provided, although it is of critical importance for appropriate technical load-flow analyses. Few datasets worldwide successfully report their exact location, technology, and fuels of each power plant, and they are often incomplete. Therefore, multiple sources shall be compared to infer the most likely mapping of power plants for any region.

The powerplantmatching tool has been successfully employed in Europe [38] and in Africa [16] for fetching multiple datasets and infer the most likely ground truth. While first tailored and tested only in Europe [38], the tool has been continuously improved, and multiple global data sources are now available. While in [16] only a handful of these datasets possess global coverage (like GEO [39], GPD [40], and IRENA [41] databases), their relevance to Africa was already acceptable, and with the recent integration of the GEM [42] database, performances have further increased. Fuel-fired generators, including coal and gas carriers, have been included, in addition to existing geothermal power plants. Optionally, capacity expansion is performed.

The strategy employed for powerplantmatching is as follows. First, raw datasets, including GEO [39], GPD [40], and GEM [42], are obtained. Then, each of them is cleaned, aggregated at power plant level, and filtered to eliminate incomplete or corrupted data. After the refining phase, the datasets are matched pairwise to identify duplicated entries, based on various factors including name of the powerplant, location, fuel, and technology. Ultimately the duplication is processed and the entries are consolidated into a singular dataset. For each power plant entry, the exact geographical location is stored, alongside the carrier, its technology and the installation data, that overall are the major input data for the PyPSA-Earth methodology developed in [16] and further improved in this paper.

2.3.3. Fossil-fuel sources

To finalize the supply side, the access to oil, gas, and coal locally and through imports is also supported. However, for the sake of simplicity and given the focus of PyPSA-Earth to model decarbonization scenarios, we decided to approximate the modelling of the exact location of existing fossil-fuel fields. Therefore, currently, no detailed input data were needed to specify the location of fossil-fuel fields.

2.4. Transmission modelling

The transport of energy carriers from one node to the other is one essential part of spatially-resolved energy system models to accurately represent the real system dynamics. To this end, the model incorporates detailed data for electricity transmission grid and natural gas networks, including both the topology and capacity. The former is inherited from the original PyPSA-Earth methodology [16] while the latter is a new feature of this study, and allows repurposing for hydrogen use.

2.4.1. Power grids

The availability of accurate power network data is crucial for constructing realistic energy system models. However, transmission system

operators rarely share open datasets of the detailed network representation to the public, including geospatial and technical data [16], and the available ones often do have different data format. To handle the complexity, several open-source packages, such as Gridkit [43], Transnet [44], and SciGrid [43], have been developed to extract networks from the available datasets, but they are limited to specific regions rather than global coverage. To bridge this gap, a novel approach has here been used to reconstruct the network topology exclusively from globally available open data [16], primarily relying on OpenStreetMap (OSM) data, which is continually updated and rich in geolocation information [45]. OSM is a collaborative project that promotes the development of georeferenced maps worldwide, including of energy and power infrastructures. In particular, information on the location of nodes and path of lines is stored, also including partial technical information, that are used to create a model-ready network representation. However, the information need further steps of cleaning and filtering.

The process of creating the power network topology used in this study involves three main steps, which have been validated in previous studies [18]: (i) downloading the raw data, (ii) filtering and cleaning the information to create model-ready data, and (iii) building a complete network representation, which includes components like transformers, substations, converters, and transmission networks (AC and DC). These steps are essential to generate accurate representations of power networks. The download step utilizes the *earth-osm* tool [46] to efficiently retrieve OSM data and extract the available information on substations, transformers, and power lines. The filtering and cleaning stage further processes the data with the goal of standardizing the entries, filling missing data, and ensuring compatibility with the PyPSA modelling framework. To enhance the quality of the OSM-extracted grid topology, an innovative approach considers a tolerance for OSM-derived coordinates. Substations located within a specified tolerance from power transmission lines with the same voltage level are merged and connected to the power lines. This prevents potential distortions in the network model due to missing data, such as short underground cables connecting the last mile of a transmission line with a substation transformer.

2.4.2. Natural gas networks

Another critical piece of infrastructure required for modelling future energy systems is the natural gas transmission network. Global datasets for gas networks, including details such as pipeline length, capacities, and precise geospatial routing, are quite scarce. Sector-coupled PyPSA-Earth utilizes two open datasets for the matter and allows the user to choose between them. The first dataset, Global Gas Infrastructure Tracker (GGIT) [47], is a global dataset that covers a substantial portion of existing networks and provides essential information such as pipeline status length, capacities, and more. Users have the flexibility to choose which pipeline statuses to include, thereby allowing the inclusion of not only existing, but also planned and proposed pipelines in the workflow and analysis. The workflow converts the capacity data to the suitable unit for the workflow when present and estimates its value from the pipeline diameter when no data exists. The second dataset, SciGRID-gas IGGIELGN [48], offers higher-detail coverage of the European continent including data about pipeline routes and sizes, it is also utilized in PyPSA-Eur [49]. In the case of this dataset, the pipeline capacities are estimated using the diameter. Additionally for both datasets, the routing of each pipeline is traced along the previously defined geospatial regions (see section *Spatial and temporal flexibility*), in order to identify which regions are interconnected and with which pipeline capacities for the clustering process. All the downloading and processing steps mentioned take place within the workflow of the model. Beyond the two datasets, richer and more detailed data could be available from private, regional, or national datasets. Therefore, the model provides users with the option to either use their own collected gas network dataset or select from two built-in datasets. Fig. 4 presents a demonstration of the feature in the United States of America showing a raw dataset fetched from GGIT and a clustered pipeline dataset produced and adapted by the model to the 48 states forming the Mainland U.S.

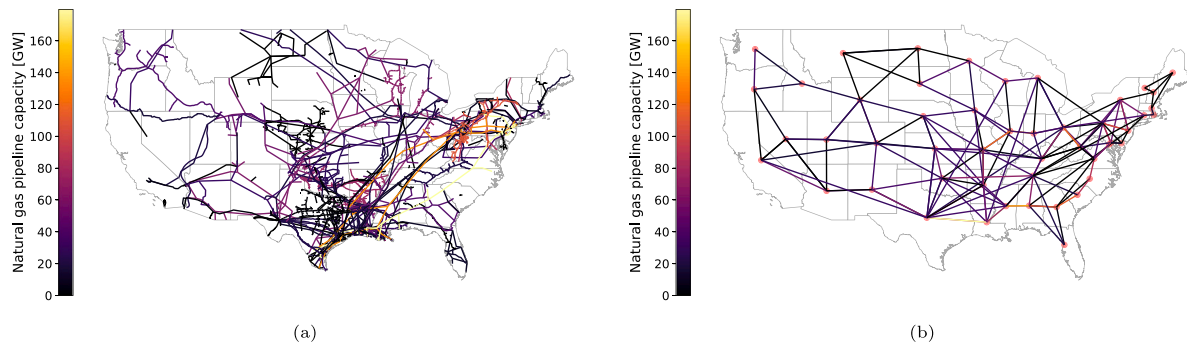


Fig. 4. Available gas pipeline data for the United States of America in sector-coupled PyPSA-Earth workflow. Figure (a) presents the raw natural gas pipeline data acquired from the GGIT dataset. Figure (b) presents the clustered gas pipeline data used in the model calculation after simplifying the network.

2.4.3. Hydrogen networks

As this study aims to model decarbonization scenarios for future energy systems, hydrogen transmission via pipelines is integrated in the model. Even though transmission-scale hydrogen networks are currently very scarce, the model allows for several options to model future networks. These options encompass both repurposing of the existing natural gas pipelines for hydrogen use and building new pipelines. Repurposing the natural gas pipelines comes at a lower capital cost [50–53] and thus is usually preferred by the system when allowed before seeking new pipeline installation. This feature completely dedicates the existing natural gas network for hydrogen use. In the case of new pipelines, the model provides two modes: using the routes of the existing natural gas pipelines leveraging existing rights-of-way and terrain advantages, or using the routes of the transmission grids as a computationally feasible proxy of a green field.

2.4.4. Other networks

In addition to electricity, natural gas and hydrogen, the model allows for the transmission of other carriers like oil and biomass, as well as for carbon dioxide. In this case, the model allows the transport with an equivalent fee that represents truck transport (biomass) or pipeline transport (oil or carbon dioxide). Due to the unavailability of data for such networks, the model allows a greenfield approach to construct such pipelines. The ability of the model to account for the transport of carbon dioxide and oil is important in several cases, for example, in the case of synthetic oil transmission from the regions of production facilities to the demand centers or for utilizing the captured carbon dioxide in one region in the synthesis process in another.

3. Model adaptability

Sector-coupled PyPSA-Earth is built to be highly adaptable to the system under consideration by the user. In particular, it adapts to the dynamic nature of the energy sector, including different demand sectors, geographical scope and temporal resolution, as well as in allowing custom data channels for user-defined data, as described in the following subsections.

3.1. Spatial and temporal flexibility

Each node of the system represents a given geospatial territory, whose size is highly customizable by the user. In particular, the model provides a flexible and adjustable way to cluster the network according to economic indicators, system infrastructure or geographic regions of interest, which are essential to dominate the model complexity. The model offers three options to cluster the system: k-means clustering, GADM clustering, and custom shape clustering. The first available approach is the k-means clustering algorithm, an optimization method that groups data points into clusters by minimizing the sum of the squared distances between each data point and the centroid of its

assigned cluster. The data points, in this case, are the geographical locations of substations, and the approach is adapted to weigh them based on their average load and conventional generation capacity. This k-means approach applications in the context of energy system modelling is explained in details in previous studies [15]. The second available approach is using the global administrative regions GADM [54] clustering method in which the model uses the shape files of the official sub-regions as the geospatial territories to cluster the substations to. Both the k-means and GADM clustering approaches are inherited from the original methodology [16]. The new third approach allows the user to provide their own custom shapefile, allowing them to precisely tailor exactly the regions they wish to model in greater detail and simplify others as needed. In this sense, the model can easily be tailored to investigate the regions of most relevance to the application in a high level of detail.

Clustering the network fundamentally means aggregating the different elements of the network in relation to the desired division forming the nodes of the network. The existing generation and demand asset capacities that geographically lie in each division are compiled as well as the renewable energy potentials and availability timeseries. The same applies to the transmission lines and pipelines whose ends connect the same geospatial regions. The model is designed to handle high temporal resolution as well by automatically rescaling the demand and supply data. The model is capable of handling hourly resolution ensuring accurate and detailed energy system representation. Both the spatial and temporal resolution configuration affect the computation time, hence the configurations should be planned accordingly. Currently, using high performance cluster that features 32 logical threads running at 4 GHz on an AMD Ryzen Threadripper PRO 5955WX processor, 512 GB of DDR4 RAM (8 × 64 GB @ 3200 MHz), and an NVIDIA T400 GPU with 4 GB of memory coupled with Gurobi [55] solver, the model computes in a reasonable amount of time (few hours up to a day) for temporal resolution of 1 h per time step up to a hundred nodes, which can be extended to a few hundred nodes with 3-h resolution. For smaller models, the model computes much faster, for example, the model to solve a network with spatial resolution 35 nodes and temporal resolution of 3-h timesteps in 35 mins on average.

3.2. Custom data channels

Even though sector-coupled PyPSA-Earth provides users with a data workflow that fully provides default data for the entire energy model, it is still valuable to modify or supplement the data sources according to their specific application. To this end, the workflow allows the user to integrate their own datasets with no programming effort. The supply, transmission, and demand data can be easily updated through the custom data feature of the model. The repository includes a guide to assist the user with the process of integrating the data and adapting it to the right format. Currently, the following datasets are easily interchangeable: renewable energy data, including the weather availability

for the generation technologies and the potential for technology specific installations, the demand data with its two steps: national-level sectoral demand data and the geolocations of the demand centers, and finally the gas transmission networks.

4. Hydrogen framework

4.1. Hydrogen export

Sector-coupled PyPSA-Earth puts special emphasis on hydrogen and incorporates special features to accommodate a wide spectrum of analyses. One notable feature is the hydrogen export option that contributes to the investigation of the future hydrogen markets. The model capitalizes on being spatially-resolved and incorporates various possible export locations per country. The total export volume is exogenously decided by the user, and the model is able to optimally select locations and timings that minimize the total system cost. The feature covers the hydrogen export supply chain until the export location within the country under study, hence, it can provide important signals about the market price of hydrogen from an integrated system before the transport cost like shipping or export pipelines.

The feature includes 4 different categories of hydrogen delivery schedules: (i) constant delivery, (ii) shipping profiles, (iii) limited flexible delivery, and (iv) totally flexible delivery. The constant hydrogen export delivery represents the export via pipeline with in-system balancing. This can include the deployment of hydrogen storage units, which buffer the time-dependent inland hydrogen generation to provide a stable hydrogen export delivery. The shipping feature integrates a temporally resolved profile as a function of export volumes, shipping capacity, travel times, and loading-unloading times. The limited flexible delivery caps the total amount of hydrogen to be delivered in a certain period of time within the year, while the totally flexible one poses no constraint on the delivery of hydrogen to the export nodes.

Since this hydrogen export in sector-coupled PyPSA-Earth is an integrated part of the larger integrated model, it allows for many valuable insights about the effect of the export on the country or region under study. This ranges from the extra cost incurred by the system and local consumers, to the necessary investments and system expansions in different locations, as presented in *Showcase A* in the next section.

4.2. Green hydrogen policies

Green hydrogen can be defined among three dimensions [56]: (i) Temporal matching, (ii) Additionality and (iii) Geographical correlation, for which sector-coupled PyPSA-Earth includes various possibilities to comply with the regulatory requirements. First, the temporal matching can be defined at yearly, monthly or hourly scale, forcing the electrical feed-in of renewable energy technologies to at least match the electricity demand of the electrolyzers in the respective time period. Second, the additionality dimension, which reflects the origin of the sourced electricity, is implicitly implemented by linking it to the temporal matching constraint. The main idea is to force the system to generate more green electricity to supply the hydrogen production. In this sense, the temporal matching constraint requires the renewable energy supply to be equal to or greater than the electricity demand of electrolyzers, hence a brownfield electricity system without renewable energies is forced to deploy renewable capacities accordingly. In systems with existing renewable energy capacities, the supply is allowed to cover the electricity demand of the electrolyzers. In this case, to avoid shifting of demand and the associated additional dispatch of fossil fuels and increase in CO₂ emissions, an additional constraint is added. The constraint accounts for the renewable generation in an identical case with no hydrogen export to ensure that the hydrogen export did not result in additional emissions. Lastly, the geographical correlation of renewable electricity supply and electrical demand of electrolyzers is set loosely (unconstrained) in sector-coupled PyPSA-Earth. More strict implementation of this dimension is planned to follow in our future work.

5. Model formulation

5.1. System representation

The various energy carriers and sectors modelled in sector-coupled PyPSA-Earth are deeply intertwined as highlighted in Fig. 5. The figure shows one representative node which consists of different carrier buses at the same location. Each carrier bus, denoted with a box (e.g. the Oil or Electricity boxes), contains generation and demand: generation is denoted with a plus (+) sign and demand is denoted with a minus (−) sign. The expandable technologies to convert energy carriers are shown as an arrow from one carrier bus to the other. This includes the electricity generation technologies that uses one of the other carriers as a fuel. For example, the open cycle gas turbines (OCGT), biomass combined heat and power plants (CHP). Further conventional generation technologies, such as nuclear, oil, coal and geothermal powerplants, are considered if already existing in the system but are not allowed to expand by default. In addition, the electricity, natural gas, or oil based heating technologies are also considered as a conversion technologies, these include heat pumps, gas and oil boilers. For the production of hydrogen, two main conversion technologies are included: electrolyzers from electricity and steam methane reforming (SMR) from natural gas. The conversion technologies also include synthetic fuel production through Fischer–Tropsch process to produce synthetic oil and two ways to produce natural gas: from hydrogen Sabatier, and from electricity, Helmeth. The available storage technologies are presented on their respective carrier buses. The connection between the different nodes of the network, is carried out through power transmission lines, hydrogen, natural gas and oil pipelines in addition to the transport of biomass and carbon dioxide. The technologies with the symbol “CC” are allow the addition of carbon capture technology. The carbon captured by those technologies can be further stored, hence completing a carbon capture and storage process (CCS), or used for other processes like Fischer–Tropsch at the same location or at another node through the carbon network explained in Section 2.4.4. For simplicity, this part has been omitted from Fig. 5. The different sectors are represented as demands on the corresponding carrier nodes. The road transport sector, for example, covers the oil, electricity (electric vehicles EV), and hydrogen (fuel-cell electric vehicles FCEV) carriers, while the industry sector is served by electricity, hydrogen, natural gas, oil, biomass and heat.

5.2. Mathematical model

This section explains the mathematical formulation of the model. First, the objective function and major mathematical constraints of the optimization model are described, then the data processing for demand assessment is detailed, and finally the extra functionalities to account for hydrogen policies and export are denoted. Table 1 provides a comprehensive list of all the abbreviations used in this section, improving the clarity of the content.

5.2.1. Core mathematical model

The core mathematical formulation of the model follows closely the work done in PyPSA-EUR [5]. Its main goal is minimizing the total annualized system cost that accounts for capital and marginal cost components. The objective function can be described using the following equation:

$$\min_{\substack{G_{n,k}, E_{n,k}, C_{n,k}, F_{l,c} \\ \dot{G}_{n,k,t}, \dot{E}_{n,k,t}, \dot{C}_{n,k,t}, \dot{F}_{l,c,t}}} \left[\overbrace{\sum_{n \in N} \sum_{k \in K} c_k G_{n,k} + \sum_{n \in N} \sum_{k \in K} c_k E_{n,k} + \sum_{n \in N} \sum_{k \in K} c_k C_{n,k} + \sum_{l \in L} c_l F_{l,c}}^{\text{annualized capital costs}} + \underbrace{\sum_{t \in T} \left[w_t \left(\sum_{n \in N} \sum_{k \in K} o_k \dot{G}_{n,k,t} + \sum_{n \in N} \sum_{k \in K} o_k \dot{E}_{n,k,t} + \sum_{n \in N} \sum_{k \in K} o_k \dot{C}_{n,k,t} + \sum_{l \in L} o_l \dot{F}_{l,c,t} \right) \right]}_{\text{marginal costs}} \right] \quad (1)$$

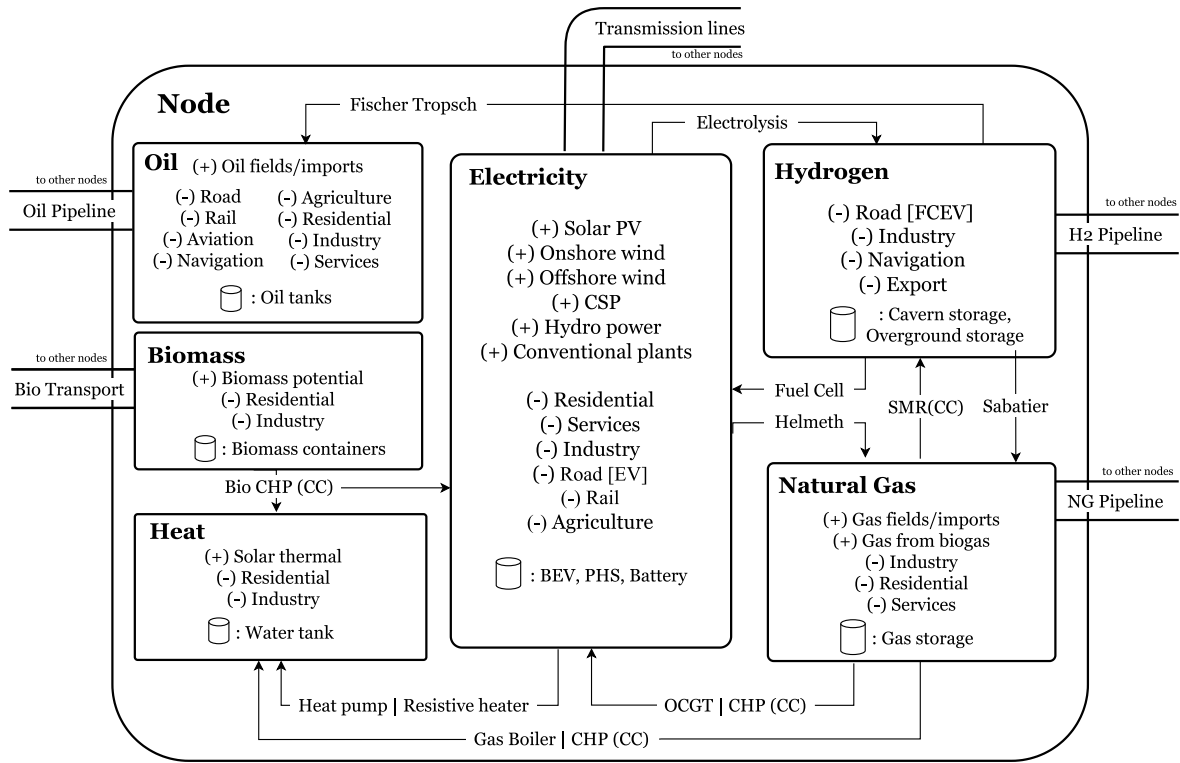


Fig. 5. Schematic of each node in the sector-coupled PyPSA-Earth model, illustrating the integration of the multiple energy carriers, including oil, biomass, heat, electricity, hydrogen, and natural gas. The residential, services, transport (navigation, aviation, road, and rail), industry, and agriculture demand sectors are represented as loads on the respective energy carrier buses. The transmission infrastructure is shown in the schematic as open-ended double links, transporting the different carriers to other nodes. A plus (+) sign indicates a source, and a minus (−) sign represents demand, while arrows depict conversion technologies that transform one carrier into another.

Table 1

Nomenclature of the model's mathematical formulation.

Indices and sets			
n, N	Index and set of network nodes ($n \in N$)	y	Year being studied
c, C	Index and set of carriers ($c \in C$)	y_b	Base year, relevant for demand projection
k, K	Index and set of technologies ($k \in K$)	H_2	Hydrogen
l, L	Index and set of inter-nodal connections ($l \in L$)	EL	Electricity
t, T	Index and set of timeframes ($t \in T$)	EY	Electrolyzer
$key_{n,s}$	Value of the distribution key at node n for sector s	ref	reference case relevant for the additionality constraint
N_x	Subset of set N representing the export nodes	RES	Renewable energy sources
w_t	Weighting factor representing the duration of the timestep		
Constants and parameters			
c_k	Annualized capital cost of technology k	$d_{n,c,s,t}$	Demand for carrier c for sector s at node n and timestep t
o_k	Operating costs of technology k	$d_{y,s,c}$	Yearly demand of carrier c for sector s in year y
c_l	Annualized capital cost of inter-nodal connection l	$d_{y_b,s,c}$	Base-case yearly demand for carrier c and sector s in reference year y_b
o_l	Operating costs of inter-nodal connection l	$d_{y,s,c,n}$	Yearly demand for carrier c and sector s at node n in year y
$\tilde{g}_s$	CAGR of sector s activity	Q_t	Time-dependent export quantity of hydrogen
$\tilde{g}_k$	CAGR of technology k efficiency gains	τ	Partial buffer hydrogen storage volume
$\tilde{e}$	Maximum capacity of the fictional hydrogen storage unit	TT	Travel time in ship profiles
$\lambda_{n,c,t}$	Shadow price of carrier c at node n and time step t	FT	Fill time in ship profiles
$\Delta f_{s,c,c}$	Fuel shift in sector s from carrier c to carrier c	PT	Pause time in ship profiles
ℓ_{CO_2}	Annual emission limit allowed	UT	Unload time in ship profiles
e_c	Emission factor of carrier c		
Decision variables			
$G_{n,k}$	Generation capacity of technology k at node n	$C_{n,k}$	Conversion capacity of technology k at node n
$\hat{G}_{n,k,t}$	Dispatch of generation technology k at node n and timestep t	$\hat{C}_{n,k,t}$	Dispatch of conversion technology k at node n and timestep t
$E_{n,k}$	Storage capacity of technology k at node n	$F_{l,c}$	Capacity of inter-nodal connection l for carrier c
$\hat{E}_{n,k,t}$	Dispatch of storage technology k at node n and timestep t	$\hat{F}_{l,c,t}$	Energy flow of carrier c through inter-nodal connection l
Q_t	Time-dependent export quantity of hydrogen	$\hat{e}_t$	Dispatch from the hydrogen fictional storage unit

where the decision variables cover both the optimal capacity of the technology and the optimal hourly dispatch. The model operates on a set of nodes N , set of carriers C , set of technologies K , set of inter-nodal connections L , and a timeframe T , normally a full year. $G_{n,k}$ denotes the optimal generation capacity of an asset of technology k at node

n and $\hat{G}_{n,k,t}$ denotes the optimal dispatch of that asset at timestep t . Similarly, the decision variables of storage, converters and energy flow are presented as $E_{n,k}$, $C_{n,k}$ and $F_{l,c}$ respectively where k uniquely identifies the technology for each category, c uniquely identifies the carrier, and l uniquely identifies the inter-nodal connections. The coefficients

represent the different cost components: c_k represents the annualized capital cost in technology k while o_k represent the operating costs. The w_t weight variable represents the duration of the timestep t and is decided by the model configuration. The main constraint in the model is the nodal balance constraint per carrier and node which ensures the satisfaction of the demand. The constraint can be represented as follows:

$$w_t \left(\sum_{k \in K} \dot{G}_{n,c,k,t} + \sum_{k \in K} \dot{E}_{n,c,k,t} + \sum_{k \in K} \dot{C}_{n,c,k,t} + \sum_{l \in L_n^{in}} \dot{F}_{l,c,t} - \sum_{l \in L_n^{out}} \dot{F}_{l,c,t} \right) = d_{n,c,s,t} \quad (\lambda_{n,c,t}), \quad \forall n \in N, c \in C, t \in T \quad (2)$$

where w_t represents the duration of the timestep t , $\dot{G}_{n,c,k,t}$ is the dispatch of generation technology k for carrier c at node n and timestep t . Similarly, $\dot{E}_{n,c,k,t}$ represents the dispatch of storage technologies and $\dot{C}_{n,c,k,t}$ denotes dispatch of the conversion technologies. $\dot{F}_{l,c,t}$ depicts the flow of carrier c through line l at timestep t , whose sign describes the direction of the flow: positive values represent flows outgoing from the origin node and incoming to the destination node, negative the converse. On the right-hand side, symbol $d_{n,c,s,t}$ denotes the demand for carrier c for sector s at node n and timestep t . The dual variable of this constraint, $\lambda_{n,c,t}$, represents the shadow price of the carrier for each node and time step, that is a reference value for the expected energy price by carrier, node and, timestep. Additionally, the total annual emissions in the system are constrained by an emission limit set by the user and can be presented as follows:

$$\sum_{t \in T} \left[w_t \left(\sum_{n \in N} \sum_{c \in C} \sum_{k \in K} e_c \dot{G}_{n,c,k,t} + \sum_{n \in N} \sum_{c \in C} \sum_{k \in K} e_c \dot{C}_{n,c,k,t} \right) \right] \leq \ell_{co_2} \quad (3)$$

where e_c is the emission factor of the specific carrier c and ℓ_{co_2} is the annual emission limit allowed. The model includes a large set of other constraints that ensure the physical feasibility of the solution, including the availability of the variable renewable energy sources, the physical limits for the generation, storage, conversion and transmission dispatch and capacities as well as the storage consistency equations as explained in the supplementary data of the sector-coupled PyPSA-EUR study [5].

5.2.2. Data processing

A novelty of the proposed method lays in the demand data workflow, which comprises several steps to populate the data to the model nodes and timesteps. The procedure aims at estimating the future demand projection using the estimated CAGR and the fuel shift values given by the user. In that sense, the total future energy demand for the country of interest can be depicted as:

$$d_{y,s,c} = d_{y_b,s,c} \cdot (1 + \dot{g}_s)^{y-y_b} \cdot (1 - \bar{g}_k)^{y-y_b} \cdot \Delta f_{s,\dot{c},c} \quad (4)$$

where $d_{y,s,c}$ denotes the demand of carrier c for sector s in year y , while $d_{y_b,s,c}$ represents the base-case demand for the reference year y_b , $\dot{g}_s$ depicts the CAGR of sector s activity. In a similar manner, $\bar{g}_k$ is the CAGR of efficiency gains of technology k . Lastly, $\Delta f_{s,\dot{c},c}$ depicts the fuel shift in sector s from carrier $\dot{c}$ to carrier c . To go from the county level demand to the nodal level, the sub-national spatial allocation of the projected sectoral demand over the nodes of the network can be depicted by the following equation:

$$d_{y,s,c,n} = \frac{key_{n,s}}{\sum_n key_{n,s}} \cdot d_{y,s,c} \quad \forall n \in N, s \in S, c \in C \quad (5)$$

where $d_{y,s,c,n}$ is the demand of carrier c for sector s in year y at node n , while $d_{y,s,c}$ is the national demand of carrier c for sector s in year y and $key_{n,s}$ is the value of the distribution key at node n for sector s . To clarify, an example of distribution key is the capacity of the industrial units for the industry sector and the population density for the residential sector.

5.2.3. Hydrogen export model

Another novelty of this paper is the hydrogen export feature. The export quantity includes a new sectoral demand for hydrogen which the network has to meet. The export demand, however, is only applied on selected nodes, called export nodes, which usually represent cross-border pipeline terminals and large scale ports suitable for ship export.

The model incorporates four different modes for hydrogen export: (i) constant delivery, (ii) ship profiles, (iii) a limitedly flexible delivery, and (iv) a totally flexible delivery. Options (i) and (ii) can be formulated as:

$$Q_t = \sum_{n \in N_x} \left[\sum_{k \in K} \dot{C}_{n,H_2,k,t} + \sum_{k \in K} \dot{E}_{n,H_2,k,t} + \sum_{l \in L_n^{in}} \dot{F}_{l,H_2,t} - \sum_{l \in L_n^{out}} \dot{F}_{l,H_2,t} - \sum_{s \in S} d_{n,H_2,s,t} \right] \quad \forall t \in T \quad (6)$$

where N_x is a subset of N representing the export nodes, Q_t is the time-dependant export quantity of hydrogen, $\dot{C}_{n,H_2,t}$ denotes all hydrogen generation conversion technologies, i.e. electrolyzers and SMR plants, $\dot{E}_{n,H_2,t}$ is the net dispatch of the hydrogen storage units at the node, $\dot{F}_{l,H_2,t}$ represents the hydrogen flow to and from the node, and $d_{n,s,H_2,t}$ is the local hydrogen demand for the different demand sectors in the node. Q_t is a hydrogen delivery timeseries that differs per export mode. For mode (i) with constant delivery, Q_t is simply defined as shown in Eq. (7), while with ship profiles, or mode (ii), it is formulated as shown in Eq. (8):

$$Q_t = \frac{Q}{|T|}, \quad \forall t \in T \quad \text{where } |T| \text{ represents the length of set } T \quad (7)$$

$$Q_t = \frac{U(t)}{\sum_T U(t)} Q, \quad \forall t \in T \quad \text{where } U(t) = \begin{cases} 1 & \text{if } t \in FT \\ 0 & \text{if } t \in TT + UT + PT \end{cases} \quad (8)$$

where FT denotes fill time which is the timesteps where the ship loads the hydrogen, TT refers to the travel time, the timesteps the ship takes to and from the destination, UT , unload time, refers to the time taken to unload the exported hydrogen and PT , pause time, is the buffer time taken. These duration values are configured by the user to suit the application. On the other hand, export modes (iii) and (iv) incorporate more elements to overcome the usual exogenous nature of demand in the model. To this end, a hypothetical hydrogen storage with no cost is added for balancing purposes. In this sense, export modes (iii) and (iv) can be formulated as:

$$Q = \sum_{t \in T} \dot{e}_t + \sum_{t \in T} \sum_{n \in N_x} \left[\sum_{k \in K} \dot{C}_{n,H_2,k,t} + \sum_{k \in K} \dot{E}_{n,H_2,k,t} + \sum_{l \in L_n^{in}} \dot{F}_{l,H_2,t} - \sum_{l \in L_n^{out}} \dot{F}_{l,H_2,t} - \sum_{s \in S} d_{n,H_2,s,t} \right] \quad (9)$$

In contrast to (6), here Q represents the total hydrogen export and the additional $\dot{e}_t$ is the dispatch from a fictional storage unit whose maximum capacity is denoted by $\bar{e}$. The capacity constraint is defined such that $\bar{e} = \tau Q$, where $\tau < 1$ for case (iii) allowing a partial buffer storage volume and $\tau = 1$ for case (iv) allowing for full flexibility.

In addition, the green hydrogen policies embodied by the temporal matching and additionality is also reflected as model constraints. The temporal matching constraint is formulated as

$$\sum_{i \in I} \sum_{n \in N} \dot{C}_{n,EL,EY,t} \leq \sum_{i \in I} \sum_{n \in N} \sum_{k \in RES} \dot{G}_{n,EL,k,t} \quad \forall i \in \tilde{T} \subset \mathcal{P}(T), \quad RES \subset K \quad (10)$$

where $\dot{C}_{n,EL,EY,t}$ represents the electricity EL input to electrolyzer EY at node n and time t and $\dot{G}_{n,EL,k,t}$ denotes the electricity dispatch of renewable energy power plants $k \in RES \subset K$ at node n and time t .

The two sides of the constraint can be matched on a yearly, monthly or timestep-basis. Formally this is represented by the configuration of the balancing intervals $\tilde{t}$. The same temporal matching constraint can also be applied with an additionality constraint. This approach utilizes a reference network with equal parameters and input data but without hydrogen export requirement in the constraint so that it reads as follows

$$\sum_{t \in \tilde{T}} \sum_{n \in N} [\dot{C}_{n,EL,EY,t}^{ref} - \dot{C}_{n,EL,EY,t}^{ref}] \leq \sum_{t \in \tilde{T}} \sum_{n \in N} \sum_{k \in RES} [\dot{G}_{n,EL,k,t} - \dot{G}_{n,EL,k,t}^{ref}] \quad \forall \tilde{t} \in \tilde{T} \subset \mathcal{P}(T), \quad RES \subset K \quad (11)$$

where $\dot{C}_{n,EL,EY,t}^{ref}$ and $\dot{G}_{n,EL,k,t}^{ref}$ represent the electrolysis input and the renewable energy power plants dispatch of the reference network, respectively. In this way, the constraint ensures that additional renewable energy electricity is generated to cover the generation of the hydrogen export required.

6. Validation

To ensure the accuracy of the model, we propose an extensive validation in this section to quantify the quality of both the data used in the workflow and its processing, thereby ensuring the reliability of the model outputs. Since the input data related to production technologies (e.g. existing powerplants, renewable sources, etc.) has been vastly validated with good results in the previous studies focusing on the power sector [16,18], in this study we focus the validation on the sector-coupled demand and transmission data. Furthermore, the model interoperability and accuracy of final results have been validated through the work done in the HYPAT project [57]. Sector-coupled PyPSA-Earth has been used as the center-piece of a model chain consisting of a total of 5 different models covering the entire supply chain of Power-to-X products export [58]. The work covers 8 countries from different regions of the world providing large spectrum of results that has been widely accepted. This not only proves that the accuracy of the model results but also further validates the model's global coverage [59].

6.1. Demand validation

The demand validation aims to assess the final energy demand data of sector-coupled PyPSA-Earth estimated with the procedure described in 2.2 relying mostly on open data. The validation is carried out for the two kinds of data used in the model: the national demand data and the geo-asset database.

The national demand validation compares the demand data in the sector-coupled PyPSA-Earth with the simplified UNSD [60] and IEA [61] databases. As mentioned in the previous sections, sector-coupled PyPSA-Earth also relies on UNSD [60] as demand data source, however this comparison ensures the quality of the internal data processing and simplification procedures. On the other hand the IEA dataset serves as a benchmark on the overall data quality, as the model has never been fed with those data; however, the IEA database is only limited to 50 counties. The comparison is performed for every combination of sector and carrier used in sector-coupled PyPSA-Earth; due to different data allocations some sectors are aggregated to match the data provided by IEA, for example the transport sector in this validation accommodate for the road, rail, aviation and navigation transport. As an example, Fig. 6 shows the comparison between the 3 datasets for Japan, showing the carrier-sector combinations relevant for the model. Overall, the figure confirms that demand data of sector-coupled PyPSA-Earth matches the UNSD and IEA data closely signalling the suitability of the proposed approach. The data accuracy slightly varies per sector and carrier, ranging between 0% in the case of residential electricity to larger than 100% as in the case of the bio fuels

for residential sector. Nonetheless, this latter has a negligible effect since the absolute value is around 2 TWh in the IEA data and 0.1 TWh in case of the UNSD data, both constituting a value well below 0.1% of the total energy demand of the country. To quantify this comparison and accommodate more weight for higher absolute values, the mean relative squared error (MRSE) is calculated for all available countries and carrier-sector combination. Table 2 shows the average MRSE for the all countries while excluding the two highest outliers for each sector-carrier combination. The processes proves the soundness of the conversion and aggregation methods applied in the workflow since the data matches closely with UNSD simplified data as all of the sectors excluding *other* show a deviation of less than 2% while for the latter it slightly increases up to a maximum of 5.8%. Also, comparing the data to the IEA energy balances shows that the approach of the two sources: IEA and UNSD match with a high degree with all values below 5% with two exceptions biomass for the *services* the *other* sectors.

In addition, similar validation processes are applied to geo-asset database of industry locations, focusing on the following main types: iron and steel, cement, chemical, and paper and pulp. Starting with iron and steel industry, the data sourced from the Global Steel Plant Tracker [26], which gathers and cross validates data from different steel organizations and governmental sources worldwide [62]. In addition to this existing validation method, we assess the data on a country basis by comparing the total capacity of the individual assets in the country to the announced total capacities from the Organisation for Economic Co-operation and Development (OECD) [63]. This is displayed for the 20 countries with highest steel capacities world wide in Fig. 7. MRSE is also employed to estimate the quality of the data and it averages to 2.2% across the 20 countries, hence highlighting the good accuracy of the approach. The same is carried out for the assets of the cement industry, with the data gathered from the spatial finance initiative by the centre for greening finance and investment (CGFI) [27] and processed with the sector-coupled PyPSA-Earth workflow. Additionally, we compare the total clinker capacity of the country for the world's largest 14 cement producer countries found in the available online source, using the United States Geological Survey (USGS) and Statista as data sources. The MRSE values are 5% and 12.6% compared to USGS and Statista data respectively, after excluding the two highest outliers; detailed results are shown in Fig. A.14. Furthermore, sector-coupled PyPSA-Earth relies on the global refinery database [29] for the chemical and petrochemical industries. The dataset is then validated twice with data available online from Concawe [64]: once against refineries in 2010 and once against the refineries in 2023. This is carried out for the 24 European countries for which the data is available from Concawe. Fig. A.15 in the annex shows how the data used in PyPSA-Earth compares to the refineries data from Concawe in 2010 and 2023. The MRSE against both years data stands at 1.9% and 10.6% after excluding the highest outlier. This proves the validity of the dataset even with small deviations compared to the current situation. For the paper and pulp industry data on the other hand, the source states that the data is still a work in progress, therefore we validated the data thoroughly on a plant level. The country level validation is done by comparing the pulp capacities to the data reported in a survey from the Food and Agriculture Organization of the United Nations (FAO) [65]. The countries assessed are limited to the Latin America countries covered in the FAO survey which are a total of 5 countries. Fig. A.13 in the annex shows the comparison between the data used in the model after we perform the plant-by-plant correction, compared to the original dataset as well as the data from the FAO survey. The plant validation we carried out enhanced the quality of the data taking it from 80.2% to 2.7% for Brazil and from greater than 100% down to 2.4% for Chile. The industries covered in sector-coupled PyPSA-Earth geo-spatial database constitute 72% of the global industry energy demand for all carriers. For the remaining industries we use the GDP distribution to disaggregate the energy demand of the different regions

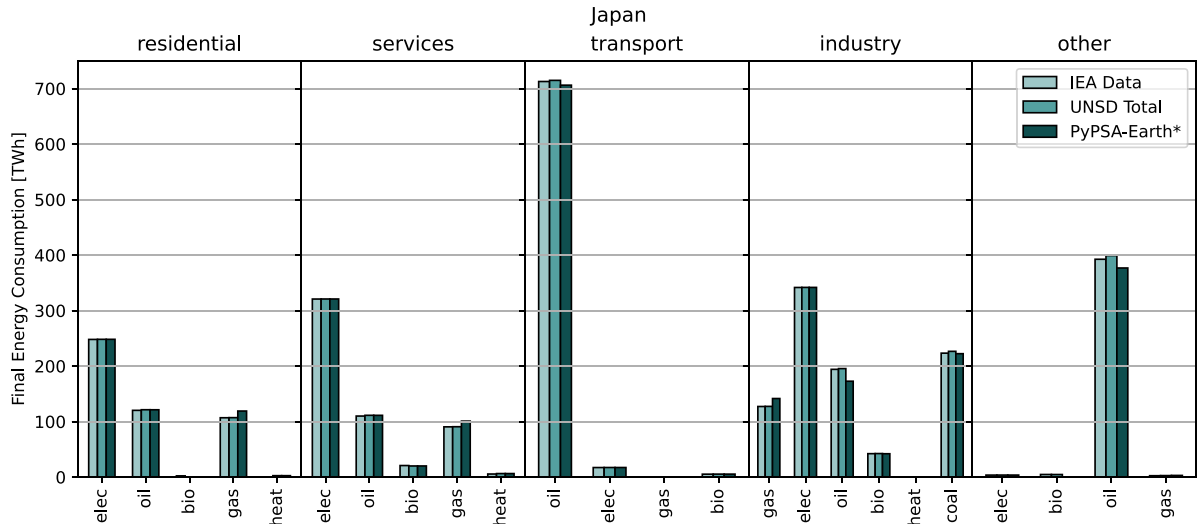


Fig. 6. Comparison of PyPSA-Earth demand data (sourced and processed from the raw UNSD data) for Japan, with UNSD [60] summary data and IEA [61] as references. The figure shows the alignment between the demand data used in the model and the source aggregation, with IEA as a reference dataset.

Table 2

Comparison of PyPSA-Earth demand data with UNSD [60] and the IEA [61]; values shown as mean relative squared error with respect to PyPSA-Earth.

Carrier	vs. UNSD total [60] [%]					vs. IEA data [61] [%]				
	Transport	Residential	Services	Industry	Other	Transport	Residential	Services	Industry	Other
Oil	0.2	0.0	0.0	0.1	5.3	0.8	1.3	0.9	2.1	3.1
Elec	0.1	0.0	0.0	0.0	2.1	2.1	0.0	0.1	0.1	5.0
Gas	0.8	1.2	1.2	1.2	3.5	2.1	2.6	3.9	1.5	4.9
Bio	0.0	0.0	0.0	0.0	5.8	0.6	3.8	8.8	1.7	22.0
Heat	–	0.0	0.0	0.0	–	–	1.6	4.5	0.3	–
Coal	–	–	–	1.9	–	–	–	–	5.4	–

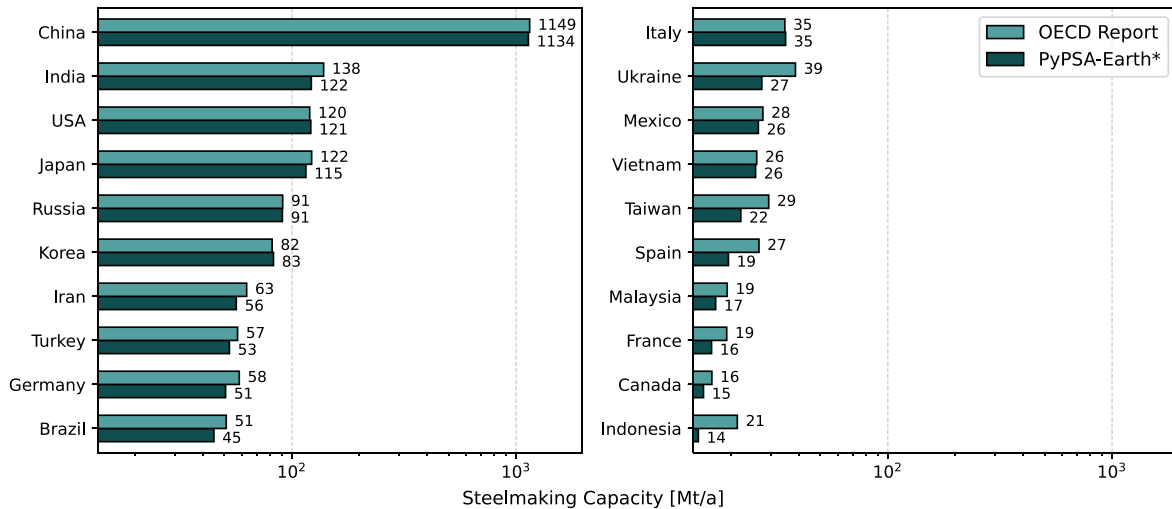


Fig. 7. Country-wise steel capacity validation shown for the 20 countries with the highest production values, highlighting the accuracy of the data used in the model for steel production capacities. The mean root squared error MRSE when comparing the PyPSA-Earth data (sourced from global steel tracker [26]) to the OECD report is 2.2%.

of a country. More exploratory and initial data analysis can be found in the sources of the UNSD data [66], the steel [67,68], and cement industries [27].

6.2. Network validation

Sector-coupled PyPSA-Earth inherits the power system model including the transmission grid from the original methodology. The

power grid data used in the model has been validated in the previous studies [16,18]. In this study we focus on validating the natural gas network topology introduced in sector-coupled PyPSA-Earth, which was missing in [16,18]. To this end we validate the two datasets included in the workflow to data from the official sources. For the GGIT network, the authors of the tool report that the data is validated in several ways including: government data and websites, reports by different companies, local non-governmental organizations tracking projects and

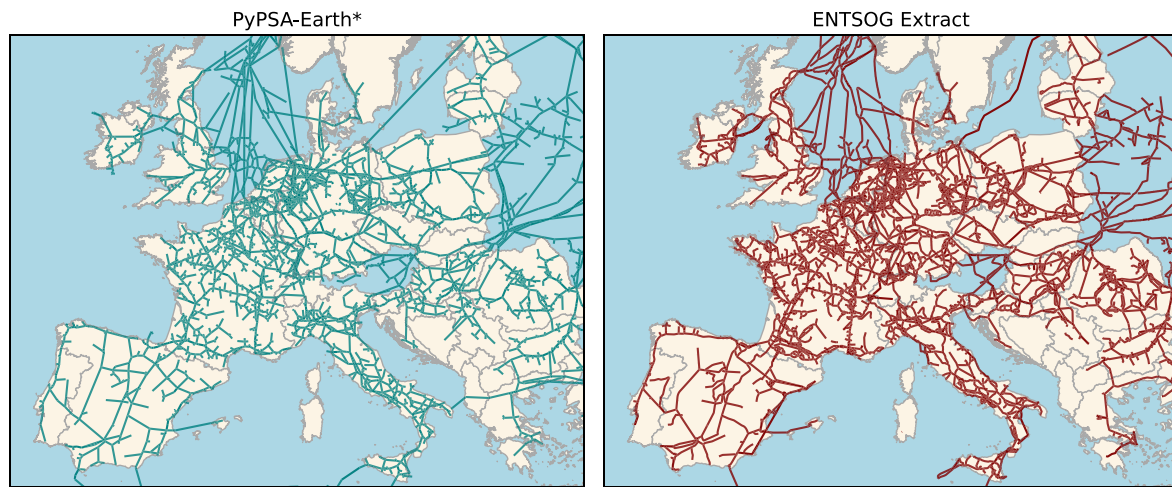


Fig. 8. Topology comparison between the pypsa-earth network dataset (sourced from IGGIELGN dataset [48]) used for the European countries in the model and the ENTSG gas network extract available through the EMAP dataset. The visual comparison proves that the topology of the used data closely aligns with the actual network topology.

on the ground contacts [69] providing as a result a wiki page for each individual pipeline with multiple data sources. Supplementing this we compare the pipeline data of the United States as an example country using the interstate pipeline data acquired from the U.S. Energy Information Administration (EIA) [70]. The GGIT and EIA datasets are shown in Fig. A.16 in the annex for visual comparison. The total length of the two networks were calculated and compared; the GGIT network 2.25 Million kilometers, and the EIA network is 2.45 Million kilometers. The GGIT network is only about 8.9% shorter than the GGIT network, indicating that they match closely in terms of overall length. The same was carried out for the IGGIELGN dataset which whose methodology and validation processes are extensively explained by the authors in the documentation files of the project [48]. The map is compared to an extract of the European Network of Transmission System Operators for Gas (ENTSG) network available through the same project in the dataset EMAP. Fig. 8 shows the two datasets spanning the European countries and visually confirming the topology match. Calculating their length also indicate a high degree of matching with only 10.9% as the IGGIELGN network has a total length of 111 thousand kilometers over the EU27 countries and the EMAP network has a total length of 124 thousand kilometres over the same countries. Further network data analysis, including exploratory and initial data analysis, can be found in the primary data source from SciGrid for the IGGIELGN dataset [48] and for the GGIT dataset from the Global Gas Tracker [71,72].

7. Showcases and results

Sector-coupled PyPSA-Earth accounts for several demand sectors and energy carriers in an integrated system that allow for a broad spectrum of cross-sectoral and cross-carrier analyses. This study primarily focuses on presenting the capabilities of the novel model; therefore the showcases have been selected to prove the regional applicability of the model as well as its features for planning decarbonized energy systems. The first one (showcase A) is a cross-carrier study case in Egypt that looks at the hydrogen export scenarios in 2030 and the resulting investment while reflecting on the effects on the electricity generation assets and natural gas use. The second (showcase B) demonstrates the ability to accommodate region-specific data in the context of a cross-sectoral planning for estimating the long-term hydrogen infrastructure for Brazil in 2050, combined to the repurposing of the natural gas grid for hydrogen use.

7.1. Showcase A: Hydrogen export potential for Egypt

In the first showcase we apply sector-coupled PyPSA-Earth on the Egyptian energy system. The country recently announced plans to export over 16% of the EU's hydrogen needs (1.667 Mtons) and capture 5% of the international market by 2030 [73]. While green hydrogen is currently seen as the primary commodity, the plan does not rule out the possibility of exporting blue hydrogen. To investigate the feasibility of such plans, the infrastructure expansion and the investments it entails are assessed using PyPSA-Earth. In this assessment the model was applied over 3 scenarios with different hydrogen production technologies allowed: (a) only electrolysis with yearly matching constraint, (b) electrolysis and SMR with CC, (c) electrolysis and SMR with and without CC.

For each scenario, the model was run once with the hydrogen export requirement of 70 TWh (approximately 2 Mtons), and once with none. Comparing the two model runs yields the additional costs incurred by the system to produce the hydrogen for export. This is presented in Fig. 9(a) highlighting the total additional costs for each scenario as well as the respective cost components. Scenario (a) with the most constraints results in the highest additional system costs with 3.68 bn€/a. The majority of the costs go to the investments in solar PV with 1.98 bn€/a financing 55 GW then electrolysis plants with 1.2 bn€/a financing 26.3 GW. The system only sees a limited investment in hydrogen pipelines as a result of the constraint set for the scenario, as no plans has been announced in the for the short-term future. In this scenario, a rather lenient hydrogen policy constraint is applied, activating a more strict one can result in a cleaner hydrogen molecules but will necessitate more investments and a higher hydrogen cost. The average cost to produce one kg of hydrogen in this scenario is 2.43 €/kg. In scenario (b) with additional system cost of 3.05 bn€/a, the dominant hydrogen production method is SMR with CC reflecting an investment of 0.87 bn€/a in 11.58 GW in addition to 1.89 bn€/a costs for natural gas used in hydrogen production, an additional investment in underground CO₂ sequestration and solar PV take place. The average hydrogen production cost in scenario (b) is 2.01 €/kg. However, this scenario needs further investigation as it requires around 100 TWh of natural gas and the country has recently been facing supply shortages [74]. For an in-depth study case, a constraint limiting the available natural gas for hydrogen generation can be easily be implemented coupled with several future scenarios. Finally, scenario (c) with no constraints on hydrogen production methods shows increase of 2.67

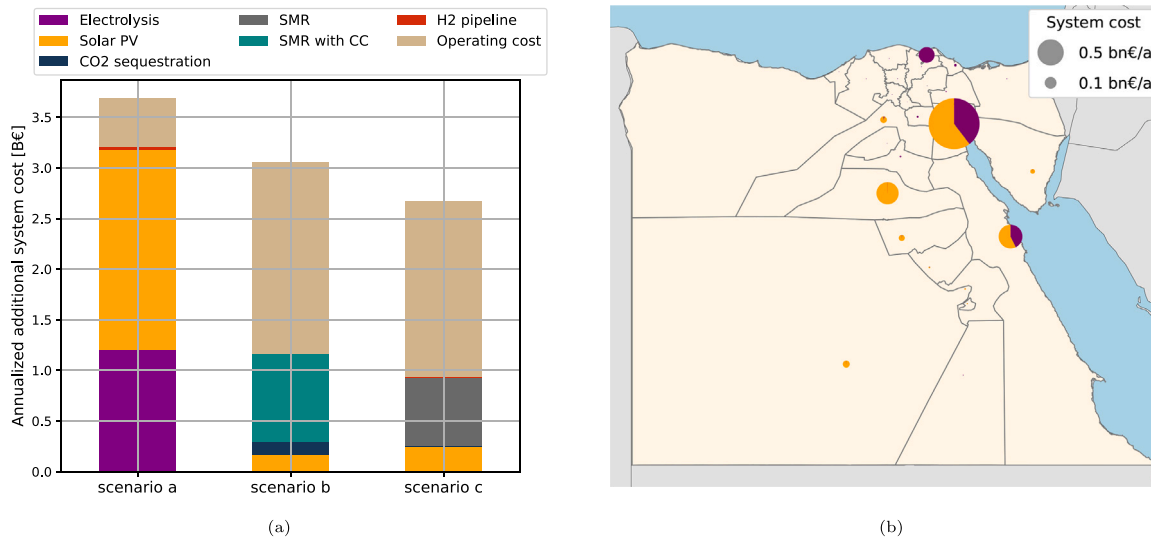


Fig. 9. Results for the hydrogen export study for Egypt: (a) shows the total annualized additional system costs broken by the cost components for the three scenarios and (b) shows the geo-spatial allocation and magnitude of optimal investments for scenario a highlighting the regions with highest infrastructure requirements.

bn€/a with most of the additional cost going to natural gas for SMR, with 1.74 bn€/a, and 0.68 bn€/a for investments in 10.49 GW of SMR. The average cost to produce one kg of hydrogen is 1.67 €/kg but this scenario only serves as a reference scenario as grey hydrogen is not an attractive commodity on the market.

The spatial distribution of the investments of scenario (a) is presented in Fig. 9(b). The largest share of investment takes place in Suez, at the same region as the first export port chosen for the showcase. The region sees huge expansion of solar PV and electrolysis, 34 GW and 17.6 GW respectively. The second and third largest electrolysis capacity expansion takes place in Damietta, the second export port in the north of the country, with capacity of 4.2 GW and Red Sea, south east of the main port of Suez with 4.1 GW. Additional solar PV expansion is seen in other regions of the country signalling an effective use of the transmission grid to supply the electrolysis plants. These findings can help quantify the effects of hydrogen export on the Egyptian system, hence demonstrating the suitability of the proposed model for decision-making on hydrogen matter. Further country-focused studies will offer even more comprehensive analysis of the financial feasibility and broader impacts on the energy infrastructure and the economic landscape.

7.2. Showcase B: Long-term hydrogen infrastructure in Brazil

While the methodology exploits by default open datasets with global scope, custom data by regional-specific authorities can be also used. In showcase B, we highlight the ability of sector-coupled PyPSA-Earth of accommodating custom region-specific data using a net-zero long-term energy infrastructure planning for Brazil in 2050.

The Brazilian Energy Research Office EPE provides, through its public portal, a selection of open data describing the structure of the Brazilian energy system at high spatial resolution [75]. Among other data, it contains the existing and planned pipelines of the natural gas transmission network, which are used to execute the proposed scenario. Before their integration into the workflow, the latter needed only to be cleaned to exclude pipelines that do not cross the boundaries of the selected cluster structure corresponding to the federal units of the country. Additionally, based on the provided diameters of the natural gas pipelines, the energy carrying capacities were calculated to ensure accurate modelling for hydrogen transport [76].

Sector-coupled PyPSA-Earth is used to estimate the needs for new power and hydrogen infrastructures, as well as for the possible conversion of existing natural gas pipelines. Opting for repurposing measures potentially proves advantageous due to their lower capital costs [50–53]. According to the brownfield approach of the default workflow, the user can choose either to use one of the two internal datasets of existing gas networks for new and repurposing of pipelines or construct new hydrogen pipelines along the routes of the current power transmission network allowing for more connected pipeline routing. In this showcase, the latter is used in case (a), whereas, in case (b) the custom dataset limits network expansion to the topology of the long-term energy infrastructure planning for Brazil. The hydrogen networks and electrolysis investments resulting from the two different cases are depicted in Figs. 10(a) and 10(b).

Results prove the flexibility of PyPSA-Earth sector-coupled in accommodating custom data and successfully suggest needs for investments in hydrogen infrastructure. In particular, hydrogen network (a) spans 12 TWkm with only new pipelines, costing 243 M€/y. Network (b) covers 17 TWkm, including 87% repurposed pipelines, with annualized investments of 186 M€/y. Reinforcements are needed in (b) between federal units in the Northeast, requiring 2 TWkm of new pipelines to distribute hydrogen output from emerging electrolysis hubs. The Northeast's federal units contribute 377 TWh in (a) and 383 TWh in (b), almost 70% of domestic production. Hydrogen must be transported to demand centers in the Southeast, comprising about half of the total demand. Network (a) connects through Pernambuco and Bahia to Minas Gerais, while (b) uses the old natural gas network along the coast, with a pipeline from Bahia to Espírito Santo surpassing (a) by 2.1 GW. Southeast's enhanced interconnection allows hydrogen supply from sites with better electrolysis conditions. In (b), regionally aggregated production is 12% lower than (a), offset by increased production in the Northeast (+6 TWh), Central-West (+6 TWh), and South (+2 TWh). Repurposing pipelines in (b) increases intraregional network density in the Southeast. Flexibility in (b) enhances cost efficiency, with annual system costs 146 M€/a lower than (a). Such insights can help guide the decision making process for planning the Brazilian transmission system, given the ongoing country interests in becoming a hydrogen hub alongside various industrial reforms. This showcase highlights a considerable difference between the two infrastructure enhancement routes. Follow-up studies can assess the exact impact of

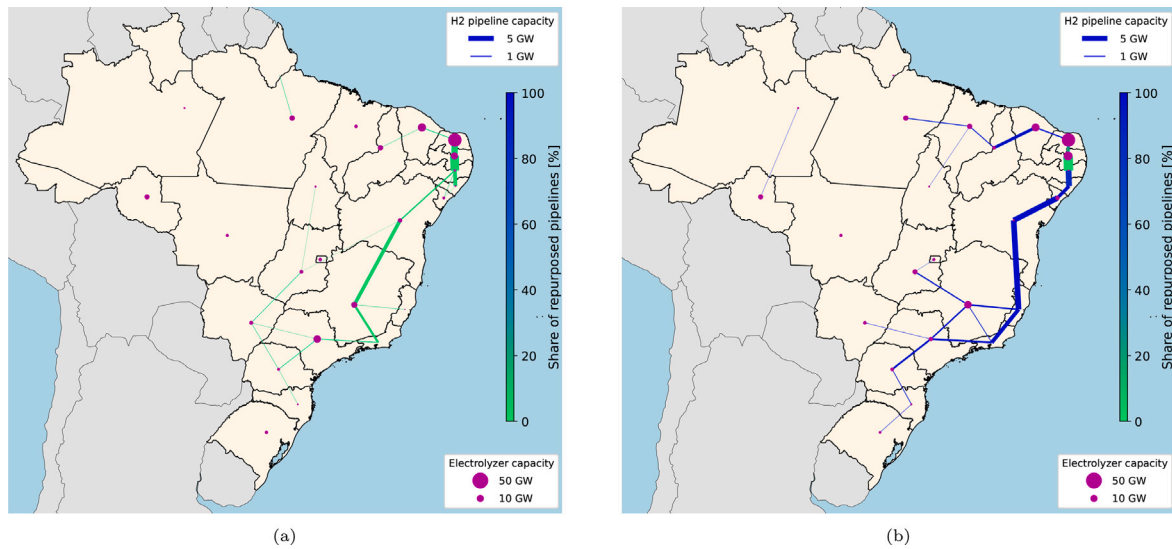


Fig. 10. Results for the hydrogen grid study for Brazil: (a) shows the pipeline and electrolysis requirements based on default pipeline routes allowed for the same routes as the transmission grid representing a greenfield proxy, and (b) shows the pipeline and electrolysis requirements based on custom pipeline routes representing the current routes of natural gas pipelines.

both routes under various future scenarios, providing valuable guidance for decision-making in the Brazilian transmission system.

8. Conclusions

This paper advances the state-of-the-art of open energy planning by presenting sector-coupled PyPSA-Earth that is a comprehensive approach to build sector-coupled multi-energy models for any country and region on Earth. By integrating major demand sectors and energy carriers at high spatial and temporal resolutions, this open-source tool facilitates policy and business decision-making for energy transition studies, including – but not limited to – capacity expansion planning, exit fossil fuel strategies, hydrogen infrastructure planning, market studies, future energy scenarios, among others. The model's innovative workflow streamlines the acquisition and processing of essential data, enabling to successfully co-optimize energy system investments and operations for various countries or regions, with good quality data below 1%–9% (excluding 1 outlier) error with respect to UNSD and IEA data.

Two showcases presented for Egypt and Brazil demonstrate that below 1–2 h the tool can successfully perform the conjoint planning of hydrogen, gas and electricity networks to satisfy electricity, thermal, and industry needs, accounting for the intertwined nature of the various energy sectors. The use of sector-coupled PyPSA-Earth for Egypt highlights that, in addition to Electrolysis, SMR with CC may be a suitable technology to meet Egypt hydrogen export targets by 2030 if the fuel is available and the market demand for blue hydrogen builds up. Further studies can build on the showcase by adding new analysis dimensions. Different future demand scenarios, utilizing the model's projection tool, can tackle the future uncertainty of different consumption patterns and technology adoption rates. Varying the carbon emission limit towards stricter values will also uncover different results in all presented scenarios. Finally, in-depth assessment of the natural gas reserves of the country will be ideal to assess the viability of SMR technology for hydrogen production and exports. The case study for Brazil, highlights instead the strong opportunities for the retrofitting of the existing natural gas transmission infrastructure to transport hydrogen within the country and also to ports for export opportunities. Further Brazil-focused studies can incorporate more details about the existing natural gas network, specifically with regards to the individual pipelines and

their suitability for retrofitting. Carrying out a sensitivity analysis for the interest rate in the country can give insights about different future economic scenarios and how this can affect the investment decisions for hydrogen transport. Finally, incorporating further industrialization scenarios, given the country's new industrial plans [77], will help in aligning the results more with the national development goals. The results presented in the showcases and the further analysis suggested in this section can be easily performed using sector-coupled PyPSA-Earth which confirms the suitability of the tool to operate across different continents and enable realistic planning studies for the countries of interest.

Future studies can accommodate improved demand modelling, as well as demand-side management schemes, and its uncertainties, which play a role in decision-making. Policy constraints, such as green hydrogen policies, will be refined to incorporate geographical considerations, while the model will be upgraded to include additional carriers like ammonia, methane, and Fischer–Tropsch products for hydrogen export. The integration of extra heat sources, such as excess heat from the industry, will further enhance the model's accuracy. The input database is expected to expand, with ongoing research into underground storage potential for carbon dioxide and hydrogen, as well as the integration of a global land-cover dataset to refine agricultural demand distribution. Detailed coverage of gas and oil fields will be considered to aid short-term analysis of countries dependent on fossil fuels. Ultimately, a major milestone will involve merging the sector-coupled version with the original PyPSA-Earth repository, providing users with the flexibility to analyse either the power system alone or the sector-coupled model.

Energy modelling and planning is of utmost importance for the energy transition, and sector-coupled PyPSA-Earth emerges as a pivotal tool in shaping comprehensive country energy systems masterplans. Its open-source framework not only fosters accessibility for all but also cultivates a culture of collaboration and knowledge sharing. The authors envision a collaborative ecosystem where stakeholders unite to minimize redundant efforts in developing energy tools and instead focus their collective expertise on addressing the multifaceted challenges of the energy transition. It is imperative to recognize that no single country can tackle these challenges in isolation; therefore, fostering global collaboration is essential for driving meaningful progress towards sustainable energy futures. PyPSA meets Earth initiative, a

global independent research community, offer a platform which is suitable for hosting such collaborations.

CRediT authorship contribution statement

Hazem Abdel-Khalek: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Leon Schumm:** Writing – original draft, Software, Project administration, Methodology, Funding acquisition, Data curation. **Eddy Jalbout:** Writing – original draft, Visualization, Software, Data curation. **Maximilian Parzen:** Writing – original draft, Software. **Caspar Schauß:** Writing – original draft, Visualization, Software, Investigation. **Davide Fioriti:** Writing – review & editing, Writing – original draft, Validation, Software, Conceptualization.

Code availability

All code is currently available on the public repository <https://github.com/pypsa-meets-earth/pypsa-earth-sec> under GPL-3 license. After the model is merged with its predecessor, the code will be available on the public repository <https://github.com/pypsa-meets-earth/pypsa-earth> under the same license. We encourage everyone interested in the research topic to participate in the model development by joining our team of contributors following the steps found on the repository.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Figs. A.1–A.16 and Tables A.1–A.3.

Table A.1
Percentage of different transportation fuels for road transport and navigation over years for the Oceania demand example presented in Section 2.

Category	Transportation fuel	2019	2030	2040	2050
Road	Internal combustion engine	100%	92%	81%	34%
	Electric vehicles	0%	7%	16%	58%
	Fuel cell electric Vehicles	0%	1%	3%	8%
Navigation	Oil ships	100%	97%	89%	74%
	Hydrogen ships	0%	3%	11%	26%

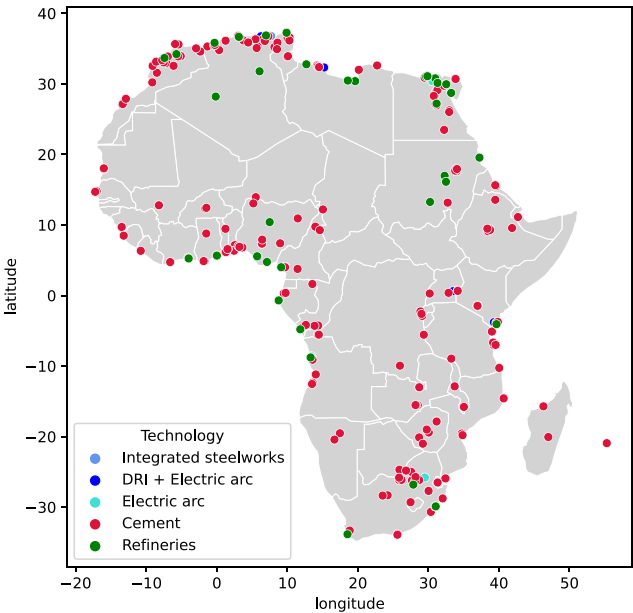


Fig. A.1. Location of major industrial sites in Africa used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

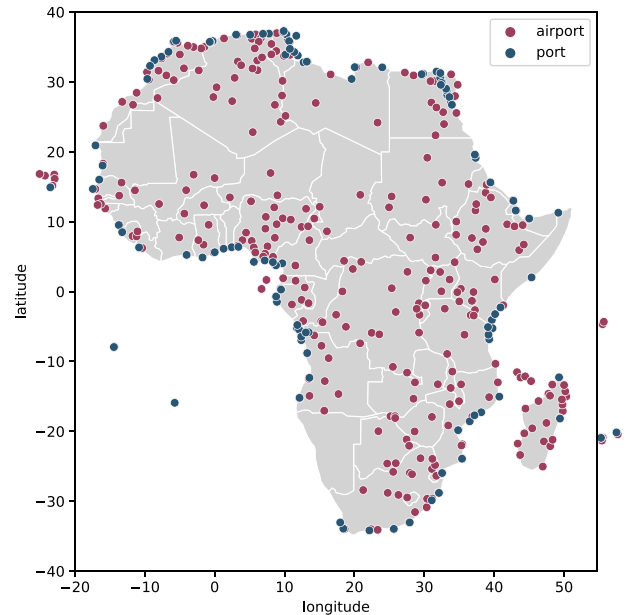


Fig. A.2. Location of major ports and airports in Africa used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

Table A.2

Compound annual growth rates CAGR of activity parameters for the different carrier demands in all sectors for the Oceania demand example presented in Section 2.

Parameter	Percentage (%)	Parameter	Percentage (%)
Residential			
Residential electricity	2.57	Residential heat (flexible)	2.31
Residential gas	1.10	Residential heat gas	0.0
Residential oil	2.57	Residential heat oil	0.0
Residential biomass	2.57	Residential heat biomass	0.0
Services			
Services electricity	2.22	Services oil	−3.58
Services gas	−0.61	Services heat (flexible)	2.00
Services biomass	0.00		
Transportation			
Total road	1.39	Total domestic aviation	1.72
Total rail	1.94	Total international aviation	2.16
Electricity rail	2.18	Total domestic navigation	0.65
		Total international navigation	2.50
Agriculture			
Agriculture electricity	2.16	Agriculture biomass	1.47
Agriculture oil	−1.86		

Table A.3

Compound annual growth rates CAGR of efficiency gains for the different carrier demands in all sectors for the Oceania demand example presented in Section 2.

Parameter	Percentage (%)	Parameter	Percentage (%)
Residential			
Residential electricity	0.27	Residential electricity heat	0.50
Residential gas	1.04	Residential heat gas	0.00
Residential oil	0.00	Residential heat oil	0.27
Residential biomass	0.50	Residential heat biomass	0.27
Services			
Services electricity	0.27	Services oil	0.00
Services gas	0.00	Services biomass	0.55
Transportation			
Road internal combustion engines	0.00	Navigation oil	0.11
Road electric vehicles	3.14	Navigation hydrogen	0.85
Fuel cell electric vehicles	2.65	Rail oil	0.50
Aviation oil	0.00	Rail electricity	0.21
Total international aviation	0.00		
Agriculture			
Agriculture electricity	0.27	Agriculture biomass	1.23
Agriculture oil	0.67		

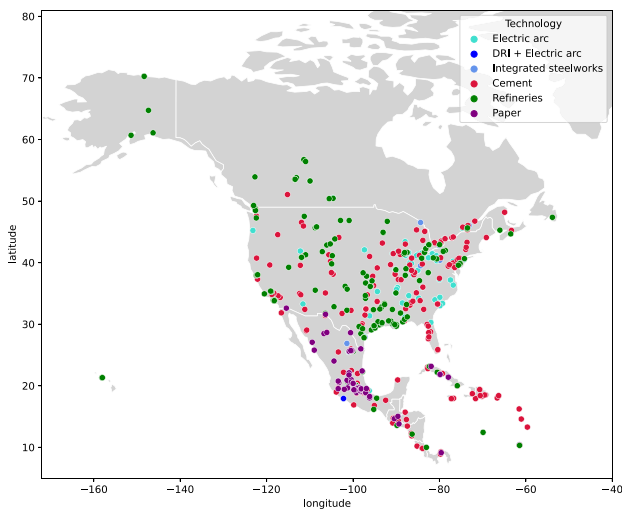


Fig. A.3. Location of major industrial sites in North America used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

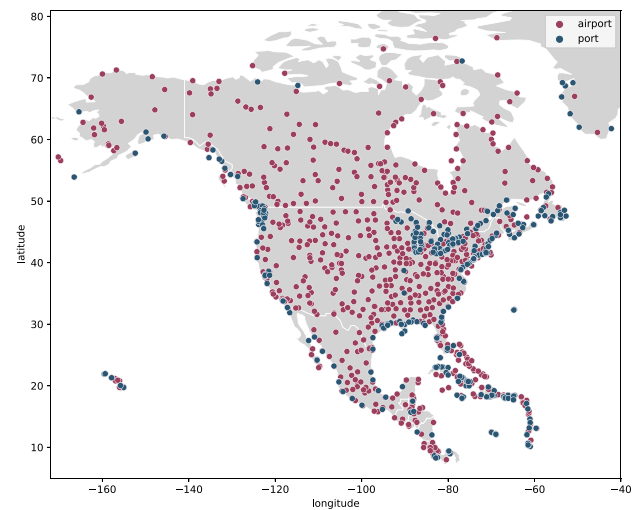


Fig. A.4. Location of major ports and airports in North America used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

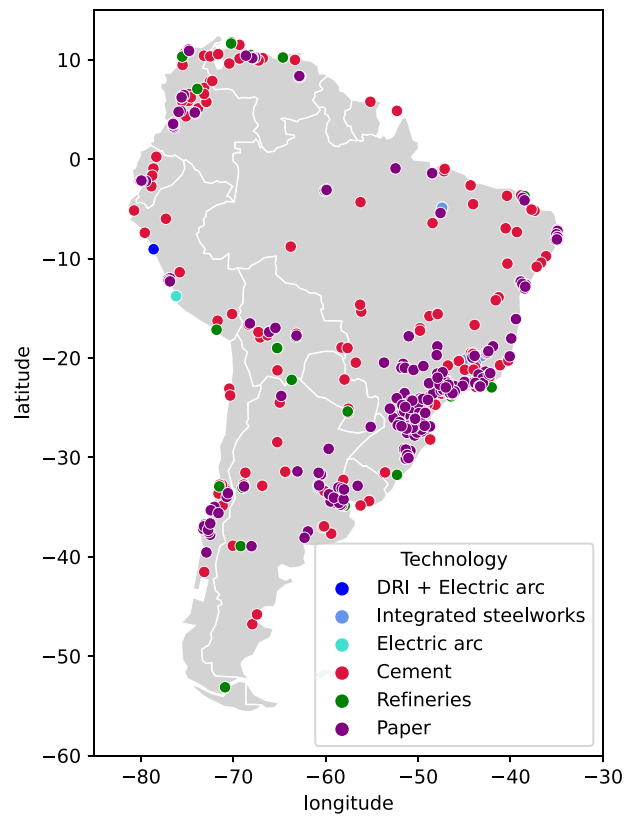


Fig. A.5. Location of major industrial sites in South America used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

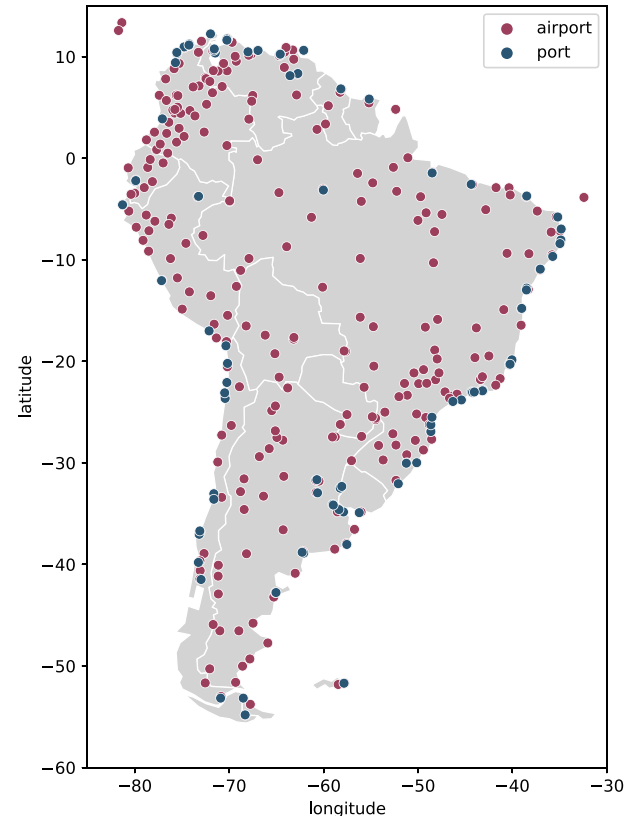


Fig. A.6. Location of major ports and airports in South America used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

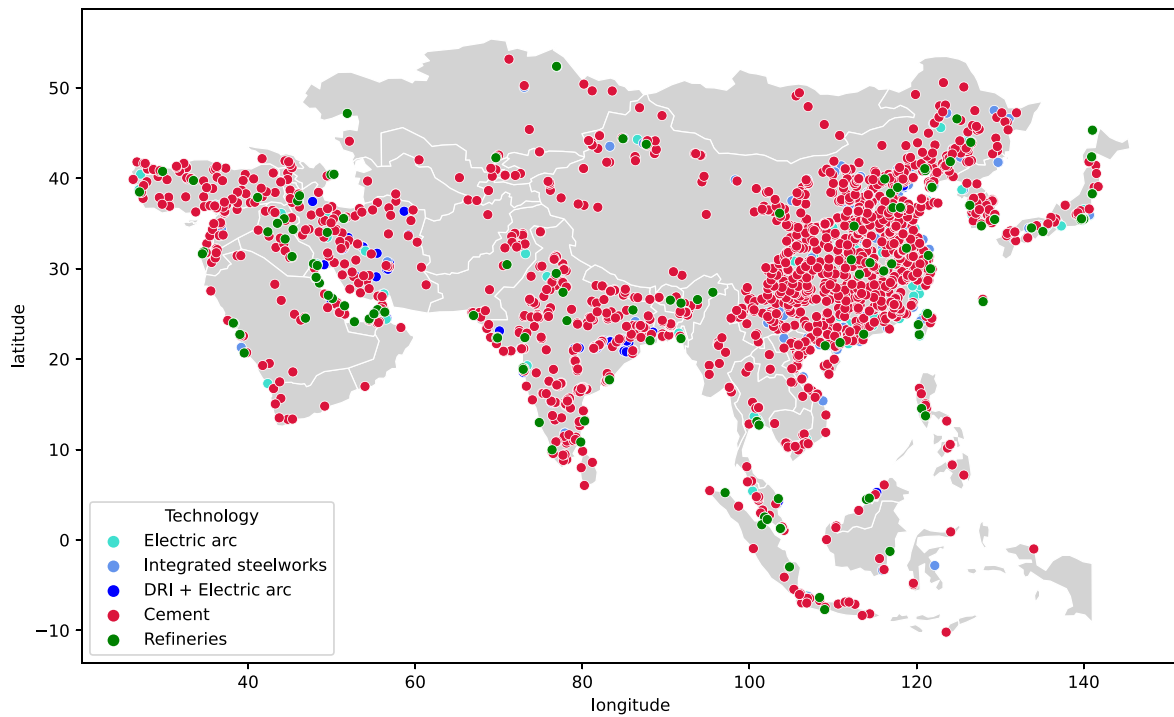


Fig. A.7. Location of major industrial sites in Asia used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

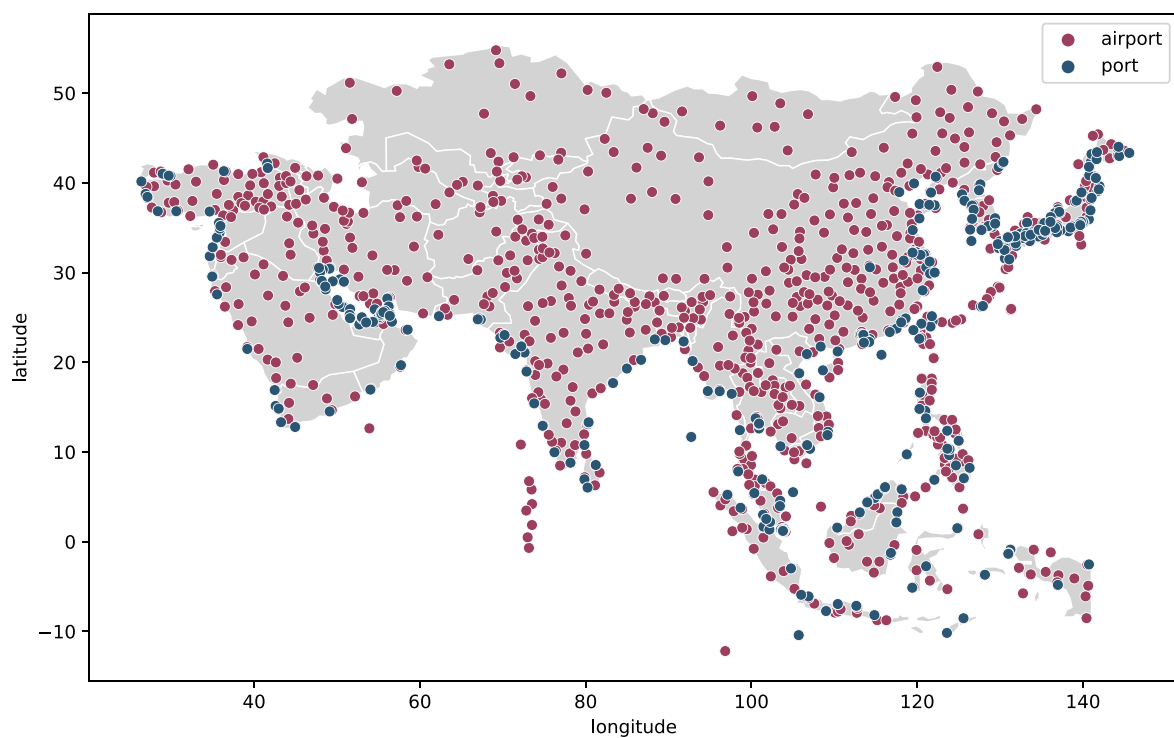


Fig. A.8. Location of major ports and airports in Asia used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

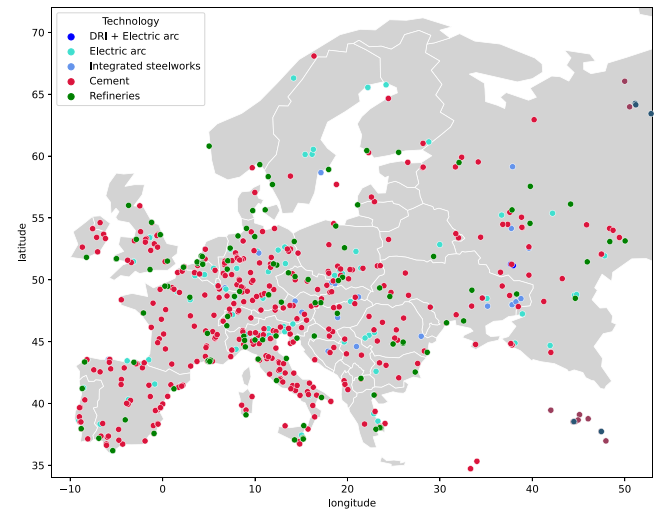


Fig. A.9. Location of major industrial sites in Europe used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

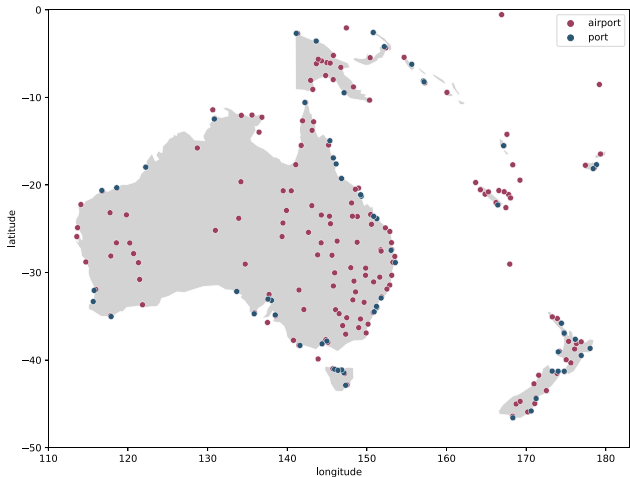


Fig. A.12. Location of major ports and airports in Oceania used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

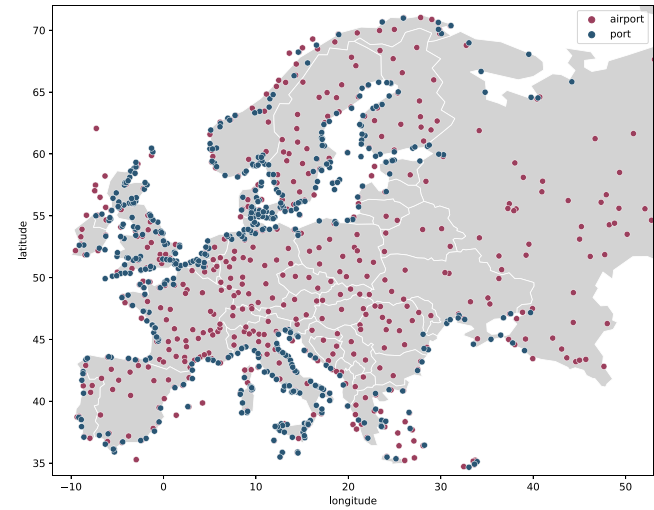


Fig. A.10. Location of major ports and airports in Europe used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

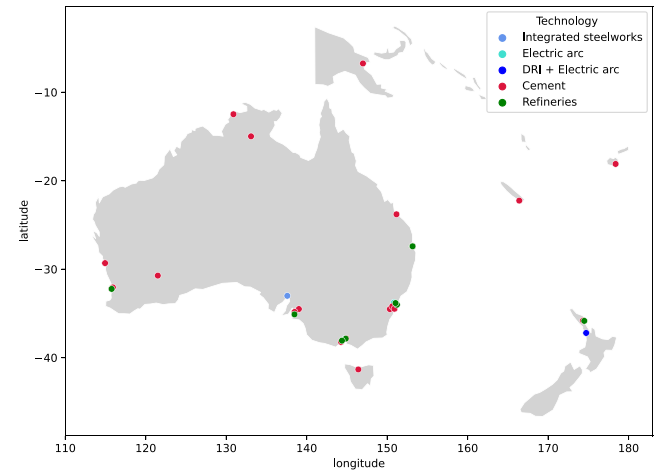


Fig. A.11. Location of major industrial sites in Oceania used mainly to disaggregate the national-level demand data into the different regions of the countries under study.

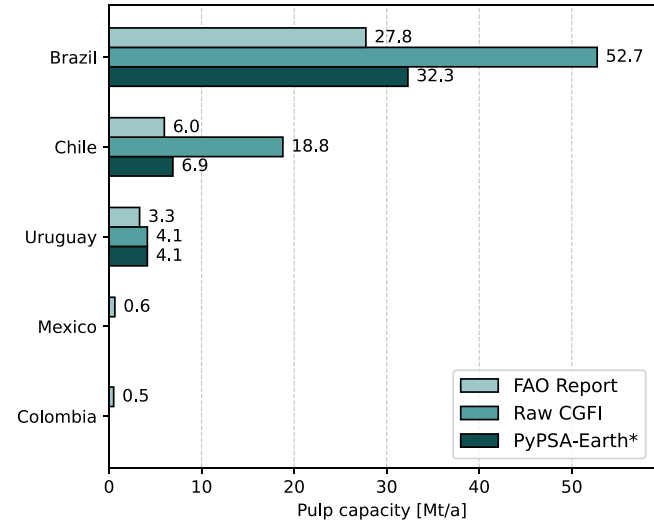


Fig. A.13. Country-wise pulp capacity validation shown for the 5 countries available through the FAO report and the CGFI data. The manual data correction enhanced the MRSE values greatly for Brazil (80.2% to 2.7%) and Chile (over 100% to 2.4%).

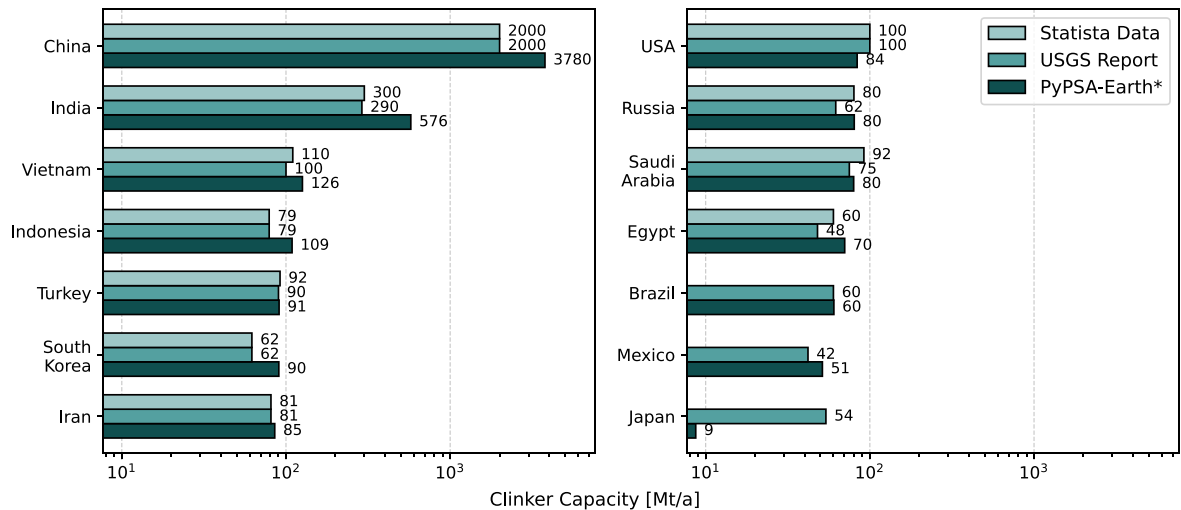


Fig. A.14. Country-wise clinker capacity validation shown for the 14 countries available through the USGS report. MRSE values stand at 5% and 12.6% compared to USGS and Statista data respectively after disregarding the two highest outliers.

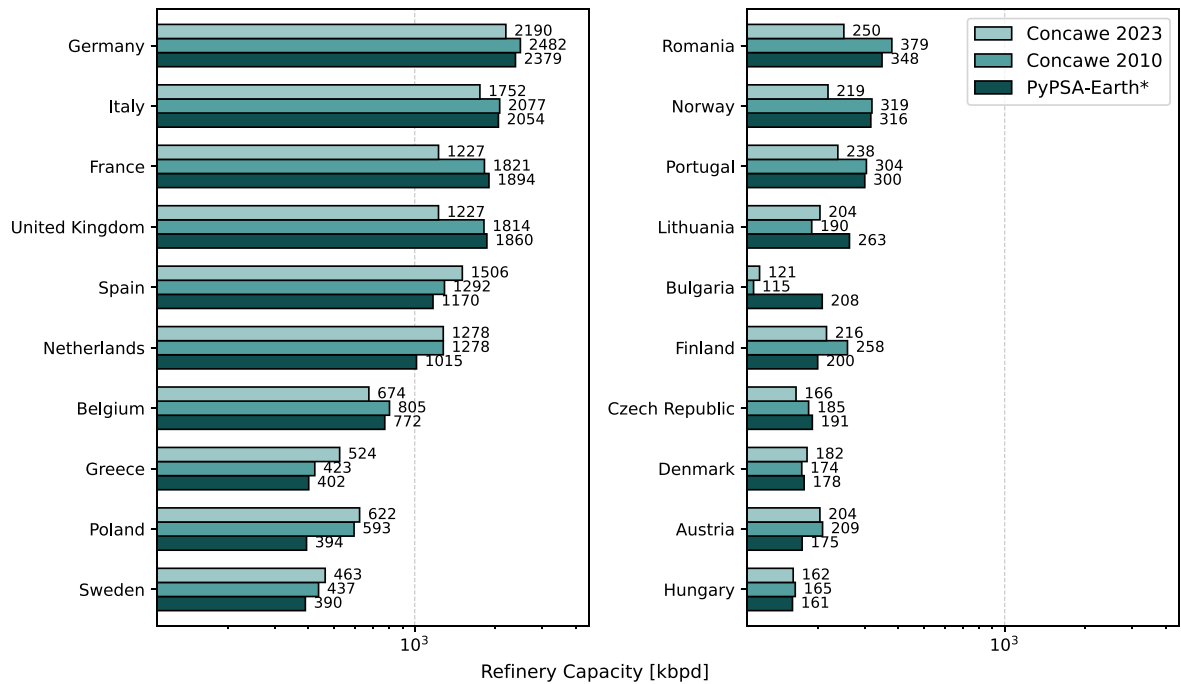


Fig. A.15. Country-wise refineries capacity validation for 20 countries available via Concawe. Comparison shows data for the years 2010 and 2023 with MRSE values of 1.9% and 10.6% after excluding respectively after disregarding the highest outlier.

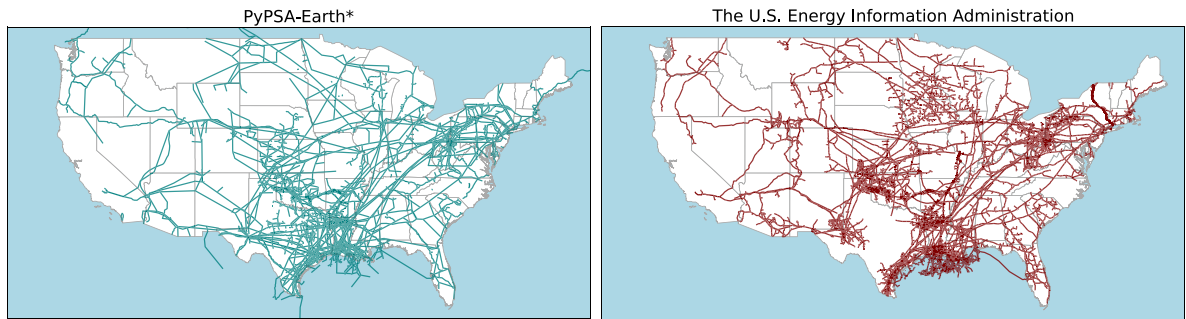


Fig. A.16. Topology comparison between the pypsa-earth network dataset in the model (sourced from GGIT dataset) and the EIA gas network extract available through official sources online. The visual comparison proves that the topology of the used data closely aligns with the actual network topology.

Data availability

The data and code used in the study can be found in open-source repository by following the steps mentioned. The link to the repository can be found in the "data availability" section.

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